while $m(\pi_n) \leq m(\pi_{n+1})$. (The details just require some inequality work and are left as Exercise 8.2.17.)

If $m(\pi_n) = M(\pi_n)$ at some stage in the process then we are done and can choose this number as I. Otherwise the intervals

$$[m(\pi_n), M(\pi_n)]$$

form a descending sequence with lengths shrinking to zero. By Cantor's intersection property (see Section 4.5.2) there is a number I so that $m(\pi_n) \to I$ and $M(\pi_n) \to I$ as $n \to \infty$. We shall show that I has the property of the theorem.

Now let $\varepsilon > 0$ and choose any partition π with the property that every interval is shorter than $\delta(\varepsilon/2)$. By what we have seen, the interval $[m(\pi), M(\pi)]$ has length smaller than $\varepsilon/2$.

Any Riemann sum over the partition π must evidently belong to the interval

$$[m(\pi), M(\pi)].$$

Let $N > 2/\varepsilon$. Suppose for a moment that the intervals

$$[m(\pi), M(\pi)] \text{ and } [m(\pi_N), M(\pi_N)]$$

intersect at some point. In that case the Riemann sum over the partition π and the value I, which is inside the interval $[m(\pi_N), M(\pi_N)]$, must be closer together than $\varepsilon/2 + 1/N$, which is smaller than ε. As this is precisely what we want to prove, we are done.

It remains to check that the two intervals

$$[m(\pi), M(\pi)] \text{ and } [m(\pi_N), M(\pi_N)]$$

intersect at some point. To find a point common to these two intervals combine the two partitions π and π_N to form a partition containing all points in either partition. The Riemann sum over such a partition belongs to the interval $[m(\pi), M(\pi)]$ and also to the interval $[m(\pi_N), M(\pi_N)]$. This completes the proof. (That I is unique is left as Exercise 8.2.2.) ■

A special case of this definition and this theorem allows us to compute an integral as a limit of a sequence. In practice this is seldom the best way to compute it, but it is interesting and useful in some parts of the theory.

Corollary 8.3: *Let f be a continuous function on an interval $[a,b]$ Then*

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=0}^{n-1} f\left(a + \frac{k}{n}(b-a)\right).$$

8.2.1 Scope of Cauchy's First Method

✂ *Enrichment section. May be omitted.*

It is natural to ask whether this method of Cauchy for describing the integral of a continuous function would apply to a larger class of functions. But Cauchy did not ask this question. His goal was to assign a meaning for continuous functions, a class of functions that was large enough for most applications. The only limitation he might have seen was that this method would fail for functions having *infinite singularities* (i.e., discontinuity points where the function is unbounded). Thus he was led to the method

we discuss in Section 8.4 as Cauchy's second method. Cauchy and other mathematicians of his time were sufficiently confused as to the meaning of the word "function" that they might never have asked such a question.

But we can. And many years later Riemann did too, as we shall see in Section 8.6. In the exercises you are asked to prove the following:

> *The first method of Cauchy will fail if applied to an unbounded function f on an interval $[a,b]$.*
>
> *The first method of Cauchy succeeds if applied to any function f that is bounded on an interval $[a,b]$ and has only finitely many discontinuities there.*

The first statement shows that the method used here to define an integral is severely limited. It can never be used for unbounded functions. Since we have restricted it here to continuous functions that is no problem; any function continuous on an interval $[a,b]$ is bounded there.

The second statement shows that the method is not, however, limited only to continuous functions even though that was Cauchy's intention. Later we will use the method to define Riemann's integral which applies to a large class of (bounded) functions that are permitted to have many, even infinitely many, points of discontinuity.

Exercises

8.2.1 To complete the computations in the introduction to this chapter, show that

$$\lim_{n\to\infty} \sum_{k=1}^{n} (1+(k)/n)^3 (1/n) = 15/4.$$

This computation alone should be enough to convince you that the definition is intended theoretically and hardly ever used to compute integrals.

SEE NOTE 226

8.2.2 Show that the number I in the statement of Theorem 8.1 is unique; that is, that there cannot be two numbers that would be assigned to the symbol $\int_a^b f(x)\,dx$.

8.2.3 If f is constant and $f(x) = \alpha$ for all x in $[a,b]$ show that

$$\int_a^b f(x)\,dx = \alpha(b-a).$$

8.2.4 If f is continuous and $f(x) \geq 0$ for all x in $[a,b]$ show that

$$\int_a^b f(x)\,dx \geq 0.$$

8.2.5 If f is continuous and $m \leq f(x) \leq M$ for all x in $[a,b]$ show that

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a).$$

8.2.6 Calculate $\int_0^1 x^p\,dx$ (for whatever values of p you can manage) by partitioning $[0,1]$ into subintervals of equal length.

8.2.7 Calculate $\int_a^b x^p\,dx$ (for whatever values of p you can manage) by partitioning $[a,b]$ into subintervals $[a,aq]$, $[aq,aq^2]$, $\ldots$, $[aq^{n-1},b]$ where $aq^n = b$. (Note that the subintervals are not of equal length, but that the lengths form a geometric progression.)

8.2.8 Use the method of the preceding exercise to show that

$$\int_1^2 \frac{dx}{x^2} = \frac{1}{2}$$

and check it by the usual calculus method.

8.2.9 Compute the Riemann sums for the integral $\int_a^b x^{-2}\,dx$ $(a>0)$ taken over a partition

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

of the interval $[a,b]$ and with associated points $\xi_i = \sqrt{x_i x_{i-1}}$. What can you conclude from this?

SEE NOTE 227

8.2.10 Compute the Riemann sums for the integral $\int_a^b x^{-1/2}\,dx$ $(a>0)$ taken over a partition

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

of the interval $[a,b]$ and with associated points

$$\xi_i = \left(\frac{\sqrt{x_i}+\sqrt{x_{i-1}}}{2}\right)^2.$$

What can you conclude from this?

8.2.11 Show that

$$\lim_{n\to\infty} n\left\{\frac{1}{(n+1)^2}+\frac{1}{(n+2)^2}+\frac{1}{(n+3)^2}+\cdots+\frac{1}{(2n)^2}\right\} = \frac{1}{2}.$$

8.2.12 Express

$$\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right)$$

as a definite integral where f is continuous on $[0,1]$.

8.2.13 Calculate

$$\lim_{n\to\infty}\frac{e^{1/n}+e^{2/n}+\cdots+e^{(n-1)/n}+e^{n/n}}{n}$$

by expressing this limit as a definite integral of some continuous function and then using calculus methods.

8.2.14 Prove that the conclusion of Theorem 8.1 is false if f is discontinuous at any point in the interval and is not bounded.

8.2.15 Prove that the conclusion of Theorem 8.1 is true if f is continuous at all but a finite number of points in the interval $[a,b]$ and is bounded.

8.2.16 Prove that the conclusion of Theorem 8.1 is true for the function f defined on the interval $[0,1]$ as follows: $f(0)=0$ and $f(x)=2^{-n}$ for each

$$2^{-n-1} < x \le 2^{-n} \qquad (n=0,1,2,3,\ldots).$$

How many points of discontinuity does f have in the interval $[0,1]$? What is the value of the number I in this case?

8.2.17 For a bounded function f and any partition π

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

of the interval $[a,b]$ write

$$M(f,\pi)=\sum_{k=1}^{n}\sup\{f(x):x\in[x_{k-1},x_k]\}(x_k-x_{k-1})$$

and

$$m(f,\pi)=\sum_{k=1}^{n}\inf\{f(x):x\in[x_{k-1},x_k]\}(x_k-x_{k-1})$$

These are called the *upper sums* and *lower sums* for the partition for the function f and were used in the proof of Theorem 8.1.

(a) Show that if π_2 contains all of the points of the partition π_1, then

$$m(f,\pi_1)\le m(f,\pi_2)\le M(f,\pi_2)\le M(f,\pi_1).$$

(b) Show that if π_1 and π_2 are arbitrary partitions and f is any bounded function, then

$$m(f,\pi_1)\le M(f,\pi_2).$$

(c) Show that if π is any arbitrary partition and f is any bounded function on $[a,b]$ then

$$c(b-a)\le m(f,\pi)\le M(f,\pi)\le C(b-a)$$

where $C=\sup f$ and $c=\inf f$.

(d) Show that with any choice of associated points the Riemann sum over a partition π is in the interval $[m(f,\pi),M(f,\pi)]$.

(e) Show that, if f is continuous, every value in the interval between $m(f,\pi)$ and $M(f,\pi)$ is equal to some particular Riemann sum over the partition π with an appropriate choice of associated points ξ_k.

(f) Show that if f is not continuous the preceding assertion may be false.

8.3 Properties of the Integral

The integral has thus far been defined just for continuous functions. We ask what properties it must have. Later we shall have to extend the scope of the integral to much broader classes of functions. It will be important to us then that the collection of elementary properties here will still be valid.

These properties exhibit the structure of the integral. They are the most vital tools to use in handling integrals both for theoretical and practical matters. Since we are restricted to continuous functions in this section, the proofs are simple. As we enlarge the scope of the integral the proofs may become more difficult, and subtle differences in assertions may arise.

Note. All functions f and g appearing in the statements are assumed to be continuous on the intervals $[a,b]$, $[b,c]$, $[a,c]$ in the statements. Thus the integrals all have meaning. This means we do not have to prove that any of these integrals exist: They do. It is the stated identity that needs to be proved in each case. To prove the identity, we consider a sequence of partitions π_n chosen so that the points in the partition are closer together

than $1/n$. Let us use the notation $S(\pi_n, f)$ to denote a Riemann sum taken over this partition for the function f with associated points chosen (say) at the left-hand endpoint of the corresponding intervals. Then

$$\lim_{n\to\infty} S(\pi_n, f) = \int_a^b f(x)\,dx.$$

We shall use this idea in the proofs.

8.4 (Additive Property) *Let f be continuous on $[a,c]$ and suppose that $a < b < c$. Then*

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx.$$

Proof. For our sequence of partitions we choose π_n to be a partition of $[a,c]$ chosen so that the points in the partition are closer together than $1/n$ and so that the point b is one of the points. Each partition π_n splits into two parts; π_n' and π_n'' where the former is a partition of $[a,b]$ and where the latter is a partition of $[b,c]$. Note that

$$S(\pi_n, f) = S(\pi_n', f) + S(\pi_n'', f)$$

by elementary arithmetic. If we let $n \to \infty$ in this identity we obtain immediately the identity in the statement we wish to prove. ■

8.5 (Linear Property) *Let f and g be continuous on $[a,b]$. Then, for all $\alpha, \beta \in \mathbb{R}$,*

$$\int_a^b [\alpha f(x) + \beta g(x)]\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

Proof. Again consider a sequence of partitions of $[a,b]$, π_n chosen so that the points in the partition are closer together than $1/n$. If $S(\pi_n, f)$ denotes a Riemann sum taken over this partition for the function f, then

$$\lim_{n\to\infty} S(\pi_n, f) = \int_a^b f(x)\,dx.$$

In the same way for g we would have

$$\lim_{n\to\infty} S(\pi_n, g) = \int_a^b g(x)\,dx.$$

But it is easy to check that

$$S(\pi_n, \alpha f + \beta g) = \alpha S(\pi_n, f) + \beta S(\pi_n, g)$$

and taking $n \to \infty$ in this identity gives exactly the statement in the property. Note that we do not have to prove that

$$S(\pi_n, \alpha f + \beta g) \to \int_a^b [\alpha f(x) + \beta g(x)]\,dx.$$

This follows from Theorem 8.1 because the integrand is continuous. ■

8.6 (Monotone Property) *Let f and g be continuous on $[a,b]$. Then, if $f(x) \le g(x)$ for all $a \le x \le b$,*

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Proof. Consider a sequence of partitions π_n chosen so that the points in the partition are closer together than $1/n$. If $S(\pi_n, f)$ denotes a Riemann sum taken over this partition for the function f, then

$$\lim_{n\to\infty} S(\pi_n, f) = \int_a^b f(x)\,dx.$$

In the same way for g we would have

$$\lim_{n\to\infty} S(\pi_n, g) = \int_a^b g(x)\,dx.$$

But since $f(x) \le g(x)$ for all x we must have

$$S(\pi_n, f) \le S(\pi_n, g).$$

Taking limits as $n \to \infty$ in this inequality yields the property. ■

8.7 (Absolute Property) *Let f be continuous on $[a,b]$. Then*

$$-\int_a^b |f(x)|\,dx \le \int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx$$

or, equivalently,

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx.$$

Proof. This follows immediately from the monotone property because

$$-|f(x)| \le f(x) \le |f(x)|.$$

■

Fundamental Theorem of Calculus The next two properties are known together as the fundamental theorem of calculus. They establish the close relationship between differentiation and integration and offer, to the calculus student, a useful method for the computation of integrals. This method reduces the computational problem of integration (i.e., computing a limit of Riemann sums) to the problem of finding an antiderivative.

8.8: (Differentiation of the Indefinite Integral) *Let f be continuous on $[a,b]$. Then the function*

$$F(x) = \int_a^x f(t)\,dt$$

has a derivative on $[a,b]$ and $F'(x) = f(x)$ at each point.

Proof. Let $h > 0$ and $x \in [a,b)$. We compute

$$F(x+h) - F(x) - hf(x) = \int_x^{x+h} (f(t) - f(x))\,dt$$

provided only that $x+h \le b$. Thus, using Exercise 8.3.1, we have

$$|F(x+h)-F(x)-hf(x)| \le h\max\{|f(t)-f(x)| : t \in [x,x+h]\}$$

and hence that

$$\left|\frac{F(x+h)-F(x)}{h} - f(x)\right| \le \max\{|f(t)-f(x)| : t \in [x,x+h]\}.$$

As f is continuous at x

$$\max\{|f(t)-f(x)| : t \in [x,x+h]\} \to 0$$

as $h \to 0+$ and this inequality shows that the right-hand derivative of F at $x \in [a,b)$ is exactly $f(x)$.

A similar argument would show that the left-hand derivative of F at $x \in (a,b]$ is exactly $f(x)$. This proves the property. ■

8.9: (Integral of a Derivative) *If the function F has a continuous derivative on $[a,b]$, then*

$$\int_a^b F'(x)\,dx = F(b)-F(a).$$

Proof. Given any $\varepsilon > 0$ there is a $\delta > 0$ so that any Riemann sum for the continuous function F' over a partition of $[a,b]$ into intervals of length less than δ is within ε of $\int_a^b F'(x)\,dx$. If

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

is such a partition then observe that, if we choose the associated points $\xi_k \in [x_{k-1},x_k]$ by the mean value theorem in such a way that

$$F(x_k)-F(x_{k-1}) = F'(\xi_k)(x_k-x_{k-1})$$

then we will have

$$F(b)-F(a) = \sum_{k=1}^{n} F(x_k)-F(x_{k-1}) = \sum_{k=1}^{n} F'(\xi_k)(x_k-x_{k-1}).$$

Since the right side of the identity is within ε of $\int_a^b F'(x)\,dx$ so too must be the value $F(b)-F(a)$. But this is true for any $\varepsilon > 0$ and hence it follows that these must be equal; that is, that

$$\int_a^b F'(x)\,dx = F(b)-F(a).$$

■

Exercises

8.3.1 If f is continuous on an interval $[a,b]$ and

$$M = \max\{|f(x)| : x \in [a,b]|\}$$

show that

$$\left|\int_a^b f(x)\,dx\right| \le M(b-a).$$

8.3.2 **(Mean Value Theorem for Integrals)** If f is continuous show that there is a point ξ in (a,b) so that

$$\int_a^b f(x)\,dx = f(\xi)(b-a).$$

8.3.3 If f is continuous and $m \le f(x) \le M$ for all x in $[a,b]$ show that

$$m\int_a^b g(x)\,dx \le \int_a^b f(x)g(x)\,dx \le M\int_a^b g(x)\,dx$$

for any continuous, nonnegative function g.

8.3.4 If f is continuous and nonnegative on an interval $[a,b]$ and

$$\int_a^b f(x)\,dx = 0$$

show that f is identically equal to zero there.

8.3.5 **(Second Mean Value Theorem for Integrals)** If f and g are continuous on an interval $[a,b]$ and g is nonnegative, show that there is a number $\xi \in (a,b)$ such that

$$\int_a^b f(x)g(x)\,dx = f(\xi)\int_a^b g(x)\,dx.$$

8.3.6 If f is continuous on an interval $[a,b]$ and

$$\int_a^b f(x)g(x)\,dx = 0$$

for every continuous function g on $[a,b]$ show that f is identically equal to zero there.

8.3.7 **(Integration by parts)** Suppose that f, g, f' and g' are continuous on $[a,b]$. Establish the *integration by parts* formula

$$\int_a^b f(x)g'(x)\,dx = [f(b)g(b) - f(a)g(a)] - \int_a^b f'(x)g(x)\,dx.$$

8.3.8 **(Integration by substitution)** State conditions on f and g so that the *integration by substitution* formula

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(s)\,ds$$

is valid.

8.3.9 State conditions on f, g and h so that the *integration by substitution* formula

$$\int_a^b f(g(h(x)))g'(h(x))h'(x)\,dx = \int_{g(h(a))}^{g(h(b))} f(s)\,ds$$

is valid.

8.3.10 If f and g are continuous on an interval $[a,b]$ show that

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \le \left(\int_a^b [f(x)]^2\,dx\right)\left(\int_a^b [g(x)]^2\,dx\right).$$

SEE NOTE 228

8.4 Cauchy's Second Method

Defining an integral only for continuous functions, as we did in the preceding section, is far too limiting. Even in the early nineteenth century the need for considering more general functions was apparent. For Cauchy this meant handling functions that have discontinuities. But Cauchy would not have felt any need to handle badly discontinuous functions, indeed he may not even have considered such objects as functions. In our terminology we could say that Cauchy was interested in extending his integral from continuous functions to functions possessing isolated discontinuities (i.e., the set of discontinuity points contains only isolated points).

We have already noted in Section 8.2.1 that *bounded* functions with finitely many discontinuities present no difficulties. Cauchy's first method can be applied to them. It is the case of *unbounded* functions that offers real resistance. What should we mean by the integral

$$\int_0^1 \frac{dx}{\sqrt{x}}?$$

While the integrand has only one discontinuity (at $x = 0$) the function is unbounded and Cauchy's first method cannot be applied. If the integral did make sense, then we would expect that the function

$$F(\delta) = \int_\delta^1 \frac{dx}{\sqrt{x}}$$

would be defined and continuous everywhere on the interval $[0,1]$ and the value $F(0)$ would equal our integral. But here $F(x)$ is not defined at $x = 0$ although it is defined for all x in $(0,1]$ since the integrand is continuous on any interval $[x,1]$ for $x > 0$. If we compute it we see that

$$F(\delta) = \int_\delta^1 \frac{dx}{\sqrt{x}} = 2 - 2\sqrt{\delta}.$$

While we cannot take $F(0)$ itself (it is not defined), we can take the limit,

$$\lim_{\delta \to 0+} F(\delta) = \lim_{\delta \to 0+} \int_\delta^1 \frac{dx}{\sqrt{x}} = 2,$$

as a perfectly reasonable value for the integral.

Indeed if we consider this as a problem in determining the area of the unbounded region in Figure 8.3 we can see graphically why the answer should be 2. Note that in the figure $0 < a < b < c < 1$ and these numbers have the values $a = 1/64$, $b = 1/16$, and $c = 1/4$. The integrals have values

$$\int_c^1 x^{-1/2}\,dx = 1, \quad \int_b^c x^{-1/2}\,dx = 1/2, \quad \text{and} \quad \int_a^b x^{-1/2}\,dx = 1/4$$

and so we would expect

$$\int_0^1 x^{-1/2}\,dx = 1 + 1/2 + 1/4 + \cdots = 2$$

as indeed this method does give.

This is precisely Cauchy's second method. If you understand this example, you understand the method. Any general write-up of the method might obscure this simple

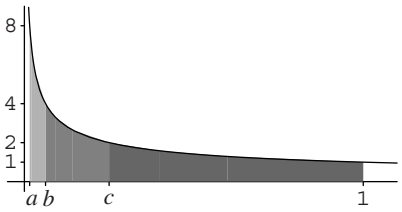


Figure 8.3. Computation of $\int_0^1 x^{-1/2}\,dx = 2$.

idea. We need some language, however. The procedure of taking a limit to obtain the final value of the integral may or may not work. If the limit does exist, we say that the integral *converges*, or is a *convergent integral*, and we say that the function f is *integrable by Cauchy's second method* or simply *integrable* if the context is clear. Otherwise the integral is said to be *divergent*.

We give a formal definition valid just in the case that the function has one point of unboundedness, and that point occurs at the left-hand endpoint of the interval. For more than one point, or for a point not at an endpoint, the definition is best generalized by splitting the integral into separate integrals, each of which can be handled one at a time in this fashion. (See Exercise 8.4.3.)

Definition 8.10: Let f be a continuous function on an interval $(a,b]$ that is unbounded in every interval $(a,a+\delta)$. Then we define

$$\int_a^b f(x)\,dx$$

to be

$$\lim_{\delta\to 0+}\int_{a+\delta}^b f(x)\,dx$$

if this limit exists, and in this case the integral is said to be *convergent*. If both integrals

$$\int_a^b f(x)\,dx \text{ and } \int_a^b |f(x)|\,dx$$

converge the integral is said to be *absolutely convergent*.

The role of the extra condition of absolute convergence is much like its role in the study of infinite series. You will recall that absolutely convergent series are more "robust" in the sense that they can be rather freely manipulated, unlike the nonabsolutely convergent series, which are rather fragile. The same is true here of absolutely convergent integrals. Note that the integral

$$\int_0^1 x^{-1/2}\,dx$$

is both convergent and absolutely convergent merely because the integrand is nonnegative.

Exercises

8.4.1 Formulate a definition of the integral $\int_a^b f(x)\,dx$ for a function continuous on $[a,b)$ and unbounded at the right-hand endpoint. Supply an example.

8.4.2 Formulate a definition of the integral $\int_a^b f(x)\,dx$ for a function continuous on $[a,c)$ and on $(c,b]$ and unbounded in every interval containing c. Supply an example.

8.4.3 How would an integral of the form

$$\int_0^3 \frac{f(x)}{\sqrt{|x(x-1)(x-2)(x-3)|}}\,dx$$

be interpreted, where f is continuous?

8.4.4 Let f and g be continuous on $(a,b]$ and such that $|f(x)| \le |g(x)|$ for all $a < x \le b$. If the integral $\int_a^b g(x)\,dx$ is absolutely convergent, show that so also is the integral $\int_a^b f(x)\,dx$.

8.4.5 For what continuous functions f must the integral

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}}\,dx$$

converge?

SEE NOTE 229

8.4.6 Let f be a bounded function, continuous on $(a,b]$ and that is discontinuous at the endpoint a. Show that if the second method of Cauchy is applied to f then the result is the same as applying the first method to the entire interval $[a,b]$ [regardless of the value assigned to $f(a)$].

8.4.7 Suppose that f is continuous on $[-1,1]$ except for an isolated discontinuity at $x = 0$. If the limit

$$\lim_{\delta \to 0+} \left(\int_{-1}^{-\delta} f(x)\,dx + \int_{\delta}^{1} f(x)\,dx \right)$$

exists does it follow that f is integrable on $[-1,1]$?

8.4.8 As a project determine which of the properties of the integral in Section 8.3 (which apply only to continuous functions on an interval $[a,b]$) can be extended to functions that are integrable by Cauchy's second method on $[a,b]$. Give proofs.

8.5 Cauchy's Second Method (Continued)

The same idea that Cauchy used to assign meaning to the integral of unbounded functions he also used to handle functions on unbounded intervals. How should we interpret the integral

$$\int_1^\infty \frac{dx}{x^2}?$$

We might try first to form a partition of the unbounded interval $[1,\infty)$ and seek some kind of limit of Riemann sums. A much simpler idea is to adapt Cauchy's second method to this in the obvious way.

$$\int_1^\infty \frac{dx}{x^2} = \lim_{X \to \infty} \int_1^X \frac{dx}{x^2} = \lim_{X \to \infty} \left(1 - \frac{1}{X}\right) = 1.$$

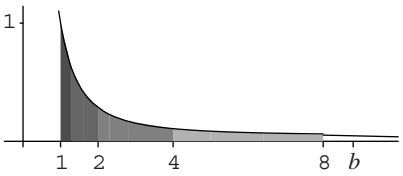


Figure 8.4. Computation of $\int_1^\infty x^{-2}\,dx = 1$.

In Figure 8.4 we show graphically how to compute the area that is represented by $\int_1^\infty x^{-2}\,dx$. Note that

$$\int_1^2 x^{-2}\,dx = 1/2,\ \int_2^4 x^{-2}\,dx = 1/4,\ \int_4^8 x^{-2}\,dx = 1/8$$

and so we would expect

$$\int_1^\infty x^{-2}\,dx = 1/2 + 1/4 + 1/8 + \cdots = 1$$

as indeed this method does give. (See Figure 8.4.)

This is precisely Cauchy's second method applied to unbounded intervals. Again, if you understand this example, you understand the method.

We give a formal definition valid just for an infinite interval of the form $[a,\infty)$. The case $(-\infty,b]$ is similar. The case $(-\infty,+\infty)$ is best split up into the sum of two integrals, from $(-\infty,a]$ and $[a,\infty)$, each of which can be handled in this fashion. (See Exercise 8.5.2.)

Definition 8.11: Let f be a continuous function on an interval $[a,\infty)$. Then we define

$$\int_a^\infty f(x)\,dx$$

to be

$$\lim_{X\to\infty}\int_a^X f(x)\,dx$$

if this limit exists, and in this case the integral is said to be *convergent*. If both integrals

$$\int_a^\infty f(x)\,dx \ \text{ and } \ \int_a^\infty |f(x)|\,dx$$

converge the integral is said to be *absolutely convergent*.

Again, the role of the extra condition of absolute convergence is much like its role in the study of infinite series. Note that the integral $\int_1^\infty x^{-2}\,dx$ is convergent and also absolutely convergent merely because the integrand is nonnegative.

Exercises

8.5.1 Formulate a definition of the integral $\int_{-\infty}^b f(x)\,dx$ for a function continuous on $(-\infty,b]$. Supply examples of convergent and divergent integrals of this type.

8.5.2 Formulate a definition of the integral $\int_{-\infty}^{\infty} f(x)\,dx$ for a function continuous on $(-\infty,\infty)$. Supply examples of convergent and divergent integrals of this type.

SEE NOTE 230

8.5.3 For what values of p is the integral $\int_1^{\infty} x^{-p}\,dx$ convergent?

8.5.4 Show that

$$\int_0^{\infty} x^n e^{-x}\,dx = n!.$$

8.5.5 Let f be a continuous function on $[1,\infty)$ such that $\lim_{x\to\infty} f(x) = \alpha$. Show that if the integral $\int_1^{\infty} f(x)\,dx$ converges, then α must be 0.

8.5.6 Let f be a continuous function on $[1,\infty)$ such that the integral $\int_1^{\infty} f(x)\,dx$ converges. Can you conclude that $\lim_{x\to\infty} f(x) = 0$?

8.5.7 Let f be a continuous, decreasing function on $[1,\infty)$. Show that the integral $\int_1^{\infty} f(x)\,dx$ converges if and only if the series $\sum_{n=1}^{\infty} f(n)$ converges.

8.5.8 Give an example of a function f continuous on $[1,\infty)$ so that the integral $\int_1^{\infty} f(x)\,dx$ converges but the series $\sum_{n=1}^{\infty} f(n)$ diverges.

8.5.9 Give an example of a function f continuous on $[1,\infty)$ so that the integral $\int_1^{\infty} f(x)\,dx$ diverges but the series $\sum_{n=1}^{\infty} f(n)$ converges.

8.5.10 Show that

$$\int_0^{\infty} \frac{\sin x}{x}\,dx$$

is convergent but not absolutely convergent.

SEE NOTE 231

8.5.11 **(Cauchy Criterion for Convergence)** Let $f : [a,\infty) \to \mathbb{R}$ be a continuous function. Show that the integral $\int_a^{\infty} f(x)\,dx$ converges if and only if for every $\varepsilon > 0$ there is a number M so that, for all $M < c < d$,

$$\left| \int_c^d f(x)\,dx \right| < \varepsilon.$$

8.5.12 **(Cauchy Criterion for Absolute Convergence)** Let $f : [a,\infty) \to \mathbb{R}$ be a continuous function. Show that the integral $\int_a^{\infty} f(x)\,dx$ converges absolutely if and only if for every $\varepsilon > 0$ there is a number M so that, for all $M < c < d$,

$$\int_c^d |f(x)|\,dx < \varepsilon.$$

8.5.13 As a project determine which of the properties of the integral in Section 8.3 (which apply only to continuous functions on a finite interval) can be extended to integrals on an infinite interval $[a,\infty]$. Give proofs.

8.6 The Riemann Integral

Thus far in our discussion of the integral we have defined the meaning of the symbol

$$\int_a^b f(x)\,dx$$

first for all continuous functions, by Cauchy's first method, and then for functions that may have a finite number of discontinuities at which the function is unbounded, by Cauchy's second method.

Let us return to Cauchy's first method. We ask just how far this method can be applied. It can be applied to all continuous functions; that was the content of Theorem 8.1. It can be applied to all bounded functions with finitely many discontinuities (Exercise 8.2.15). It can be applied to *some* bounded functions with infinitely many discontinuities (Exercise 8.2.16).

Rather than search for broader classes of functions to which this method applies, we adopt the viewpoint that was taken by Riemann. We simply *define* the class of all functions to which Cauchy's first method can be applied and then seek to characterize that class. This represents a much more modern point of view than Cauchy would have taken with his much more limited idea of what a function is. Note that we need only turn Theorem 8.1 into a definition.

Definition 8.12: Let f be a function on an interval $[a,b]$. Suppose that there is a number I such that for all $\varepsilon > 0$ there is a $\delta > 0$ so that

$$\left| \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) - I \right| < \varepsilon$$

whenever $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$, is a partition of the interval $[a,b]$ into subintervals of length less than δ and each ξ_k is a point in the interval $[x_{k-1},x_k]$. Then f is said to be *Riemann integrable* on $[a,b]$ and we write

$$\int_a^b f(x)\,dx$$

for that number I.

We can call the set of points

$$\pi = \{x_0,x_1,x_2,\ldots,x_{n-1},x_n\}$$

the *partition* of the interval $[a,b]$ or, equivalently, if it is more convenient, the set of intervals

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

can be called the partition. The points ξ_k that are chosen from each interval $[x_{k-1},x_k]$ are called the *associated points* of the partition. Notice that in the definition the associated points can be freely chosen inside the intervals of the partition.

Loosely a function f is Riemann integrable if the limit of the Riemann sums for f exists over that interval. The program now is to determine what classes of functions are Riemann integrable and to obtain characterizations of Riemann integrability. We shall investigate this in the remainder of this section.

We need also to find out whether the properties of the integral that hold for continuous functions now continue to hold for all Riemann integrable functions. We shall consider that in the next section.

Two observations are immediate from our earlier work:

All continuous functions are Riemann integrable.

All Riemann integrable functions are bounded.

In light of this last statement we see that the Riemann integral is somewhat limited in that it will not do anything to handle unbounded functions. For that we must still return to Cauchy's second method. But, as we shall see, the Riemann integral will handle many bounded functions that are badly discontinuous (but not too badly). As research progressed in the nineteenth century the Riemann integral became the standard tool for discussing integrals of bounded functions. For unbounded functions Cauchy's second method continued to be employed, although other methods emerged.

By the early twentieth century the Riemann integral was abandoned by all serious analysts in favor of Lebesgue's integral. The Riemann integral survives in texts such as this mainly because of the technical difficulties of Lebesgue's better, but more difficult, methods.

8.6.1 Some Examples

All Riemann integrable functions are bounded. All continuous functions are Riemann integrable. In order to obtain some insight into the question as to what functions are Riemann integrable we present some examples, first of a bounded function that is not integrable and then of a badly discontinuous function that is integrable.

Example 8.13: Here is an example of a function that is bounded but "too discontinuous" to be Riemann integrable. On the interval $[0,1]$ let f be the function equal to 1 for x rational and to 0 for x irrational. Let

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

be any partition. If we choose associated points $\xi_k \in [x_{k-1},x_k]$ so that ξ_k is rational, then the Riemann sum

$$\sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) = \sum_{k=1}^{n} (x_k - x_{k-1}) = 1 \tag{2}$$

while if we choose associated points $\eta_k \in [x_{k-1},x_k]$ so that η_k is irrational

$$\sum_{k=1}^{n} f(\eta_k)(x_k - x_{k-1}) = 0. \tag{3}$$

Because of (2) and (3), the integral $\int_0^1 f(x)\,dx$ cannot exist. ◀

Example 8.14: Recall the Dirichlet function (Section 5.2.6) which provides an example of a function that is discontinuous at every rational number and continuous at every irrational. We show that this function is Riemann integrable. On the interval $[0,1]$ let $f(0) = 0$, at rational numbers $f(p/q) = 1/q$ for $0 < p/q \leq 1$ (assuming that p/q has been expressed in its lowest terms) and to $f(x) = 0$ for x irrational.

Let $\varepsilon > 0$. Let q_0 be any positive integer larger than $2/\varepsilon$. We count the number of points x in $[0,1]$ at which $f(x) > 1/q_0$. There are finitely many of these, say M of them.

Choose δ_1 sufficiently small so that any two of these points are further apart than $2\delta_1$. Choose $\delta < \delta_1$ so that (for reasons that become clear only after all our computations are done) $M\delta < \varepsilon/2$. This will allow us to use the inequality

$$M\delta + 1/q_0 < \varepsilon. \tag{4}$$

Let

$$[x_0, x_1],\ [x_1, x_2],\ \ldots,\ [x_{n-1}, x_n]$$

be any partition chosen so that each of the intervals is shorter than δ. For any choice of associated points $\xi_k \in [x_{k-1}, x_k]$ we note that either

(i) $f(\xi_k) = 0$ if $\xi_k = 0$ or if ξ_k is irrational, or

(ii) $f(\xi_k) > 1/q_0$ if ξ_k is one of the M points counted previously, or

(iii) $0 < f(\xi_k) \leq 1/q_0$ if ξ_k is any other rational point.

We can estimate the Riemann sum

$$\sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1})$$

by considering separately these three cases. Case (i) evidently contributes nothing to this sum. Case (ii) can contribute at most $M\delta$ to this sum since each interval in the partition can contain at most one of the points of type (ii) and there are only M such points. Finally, case (iii) can contribute in total no more than $1/q_0$. Thus, using the inequality (4), we have

$$0 \leq \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) \leq M\delta + 1/q_0 < \varepsilon.$$

This proves that the integral $\int_0^1 f(x)\,dx = 0$. Considering just how discontinuous this function is (it has a dense set of discontinuities) it is startling that it is nonetheless integrable. ◀

8.6.2 Riemann's Criteria

✂ *Advanced section. May be omitted.*

What bounded functions then are Riemann integrable? The answer is that such functions must be "mostly" continuous. The example of the very discontinuous function in Example 8.13 suggests this. On the other hand, Example 8.14 shows that the discontinuities of a Riemann integrable function might even be dense. Riemann first analyzed this by using the oscillation of the function f on an interval. We recall (Definition 6.24) that this is defined as

$$\omega f([c,d]) = \sup_{x\in[c,d]} f(x) - \inf_{x\in[c,d]} f(x).$$

This measures how much the function f changes in the interval $[c,d]$. For a continuous function this is just the difference between the maximum and minimum values of f on $[c,d]$ and will be small if the interval $[c,d]$ is small enough.

Theorem 8.15 (Riemann) *A function f defined on an interval $[a,b]$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ so that*

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon$$

whenever $[x_0,x_1]$, $[x_1,x_2]$, ..., $[x_{n-1},x_n]$, is a partition of the interval $[a,b]$ into subintervals of length less than δ.

Proof. If f is Riemann integrable on $[a,b]$ with integral I, then for any $\varepsilon > 0$ there must be a $\delta > 0$ so that any two Riemann sums taken over a partition with intervals smaller than δ are both within $\varepsilon/4$ of I. In particular, we have

$$\left| \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1}) - \sum_{k=1}^{n} f(\eta_k)(x_k - x_{k-1}) \right| < \varepsilon/2$$

whenever $[x_0,x_1]$, $[x_1,x_2]$, ..., $[x_{n-1},x_n]$, is a partition of the interval $[a,b]$ into subintervals of length less than δ. Here ξ_k and η_k are any choices from $[x_{k-1},x_k]$. We rewrite this as

$$\left| \sum_{k=1}^{n} [f(\xi_k) - f(\eta_k)](x_k - x_{k-1}) \right| \leq \varepsilon/2 < \varepsilon. \tag{5}$$

Now notice that

$$\sup_{\eta,\xi \in [x_{k-1},x_k]} (f(\xi) - f(\eta)) = \omega f([x_{k-1},x_k]).$$

Thus we see that the criterion follows immediately on taking sups over these choices of ξ_k and η_k in the inequality (5).

The other direction of the theorem can be interpreted as a "Cauchy criterion" and proved in a manner similar to all our other Cauchy criteria so far in the text (indeed similar to the proof of Theorem 8.1). We omit the details. ■

Theorem 8.15 offers an interesting necessary and sufficient condition for integrability. It is awkward to use the sufficiency criterion here since it demands that we check that *all* small partitions have a certain property. The following variant is a little easier to apply since we need find only *one* partition for each positive ε.

Theorem 8.16: *A function f on an interval $[a,b]$ is Riemann integrable if and only if for every $\varepsilon > 0$ there is at least one partition $[x_0,x_1]$, $[x_1,x_2]$, ..., $[x_{n-1},x_n]$, of the interval $[a,b]$ so that*

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon.$$

Proof. By Theorem 8.15 we see that if f is Riemann integrable there would have to exist such a partition.

In the opposite direction we must show that the condition here implies integrability. Certainly this condition implies that f is bounded (or else this sum would be infinite) and so we may assume that $|f(x)| \leq M$ for all x. This gives us a useful, if crude, estimate on the size of the oscillation on any interval $[c,d]$:

$$\omega f([c,d]) \leq 2M.$$

Let $\varepsilon > 0$. We shall find a number δ so that the criterion of Theorem 8.15 is satisfied. Let $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$ be the partition whose existence is given. We use that to find our δ. Choose δ sufficiently small so that

$$2Mn\delta < \varepsilon.$$

Now let

$$[y_0,y_1],\ [y_1,y_2],\ \ldots,\ [y_{m-1},y_m],$$

be any partition of the interval $[a,b]$ into subintervals of length less than δ. These intervals are of two types: Type (i) are those that are contained entirely inside intervals of our original partition, and type (ii) are those that include as interior points one of the points x_k for $k = 1,2,\ldots,n-1$. In any case there are only $n-1$ of these intervals and each is of length less than δ. Thus, using just a crude estimate on each of these terms, the intervals of type (ii) contribute to the sum

$$\sum_{k=1}^{m} \omega f([y_{k-1},y_k])(y_k - y_{k-1})$$

no more than $(2M)n\delta$. The sum taken over all the type (i) intervals must be smaller than

$$\sum_{k=1}^{m} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon.$$

Thus the total sum

$$\sum_{k=1}^{m} \omega f([y_{k-1},y_k])(y_k - y_{k-1}) < 2Mn\delta + \varepsilon < 2\varepsilon.$$

It follows by the criterion in Theorem 8.15 that f is Riemann integrable as required. ■

8.6.3 Lebesgue's Criterion

✂ *Advanced section. May be omitted.*

Theorem 8.16 is beautiful and seemingly characterizes the class of Riemann integrable functions in a meaningful way. But at the time of Riemann there was only an imperfect understanding of sets of real numbers and so it did not occur to Riemann that the property of Riemann integrability for a bounded function f depended exclusively on the nature of the set of points of discontinuity of f. Indeed the condition

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon$$

on the oscillation of the function suggests that something more subtle than just this is happening.

In 1901 Henri Lebesgue completed this theorem by using the notion of a set of measure zero. Recall (from Section 6.8) that a set E of real numbers is of measure zero if for every $\varepsilon > 0$ there is a sequence of intervals $\{(c_i,d_i)\}$ covering all points of E and with total length $\sum_{i=1}^{\infty}(d_i - c_i) < \varepsilon$. The exact characterization of Riemann integrable functions is precisely this: They are bounded (as we already well know) and they are

continuous at all points except perhaps at the points of a set of measure zero. (In modern language they are said to be continuous *almost everywhere* or continuous a.e.)

Theorem 8.17 (Riemann-Lebesgue) *A function f on an interval $[a,b]$ is Riemann integrable if and only if f is bounded and the set of points in $[a,b]$ at which f is not continuous is a set of measure zero.*

Proof. The necessity is not difficult to prove but is the least important part for us. The sufficiency is more important and harder to prove. Throughout the proof we require a familiarity with the notion of the oscillation $\omega_f(x)$ of a function f at a point x as discussed in Section 6.7. Recall that this value is positive if and only if f is discontinuous at x.

Let us suppose that f is Riemann integrable. Certainly f is bounded. Fix $e > 0$ and consider the set $N(e)$ of points x such that the oscillation of f at x is greater than e; that is, so that

$$\omega_f(x) > e.$$

Any interval (c,d) that contains a point $x \in N(e)$ will certainly have

$$\omega f([c,d]) \geq e.$$

Let $\varepsilon > 0$ and use Theorem 8.15 to find intervals

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n],$$

forming a partition of the interval $[a,b]$ and such that

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k](x_k - x_{k-1}) < \varepsilon e/2.$$

Select from this collection just those intervals that contain a point from $N(e)$ in their interior. The total length of these intervals cannot exceed $(e\varepsilon)/(2e)$ since for each such interval $[x_{k-1},x_k]$ we must have $\omega f([x_{k-1},x_k]) \geq e$.

Thus we have succeeded in covering the set $N(e)$ by a sequence of open intervals (x_{k-1},x_k) of total length less than $\varepsilon/2$, except for an oversight. One or more of the points $\{x_i\}$ might be in the set $N(e)$, and we have neglected to cover it. Since there are only finitely many such points, we can add a few sufficiently short intervals to our collection.

Thus we have proved that for any $\varepsilon > 0$ the set $N(e)$ can be covered by a collection of open intervals of total length less than ε. It follows that $N(e)$ has measure zero. But the set of points of discontinuity of f is the union of the sets $N(1)$, $N(1/2)$, $N(1/4)$, $N(1/8)$, ... Since each of these is measure zero, it follows from Theorem 6.34 that the set of points of discontinuity of f has measure zero too as required.

This proves the theorem in one direction. In the other suppose that f is bounded, say that $|f(x)| \leq M$ for all x and that the set E of points in $[a,b]$ at which f is not continuous is a set of measure zero. Let $\varepsilon > 0$. By Theorem 8.16 we need to find at least one partition

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n]$$

of the interval $[a,b]$ so that

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon.$$

Let E_1 denote the set of points x in $[a,b]$ at which the oscillation is greater than or equal to $\varepsilon/(2(b-a))$; that is, for which

$$\omega_f(x) \geq \varepsilon/(2(b-a)).$$

This set is closed (see Theorem 6.27) and, being a subset of E, it must have measure zero. Now closed sets of measure zero can be covered by a *finite* number of small open intervals of total length smaller than

$$\varepsilon/(4M+1).$$

(See Theorem 6.35.) We can assume that these open intervals do not have endpoints in common. Note that, at points in the intervals that remain, the oscillation of f is smaller than $\varepsilon/(2(b-a))$. Consequently, these intervals may be subdivided into smaller intervals on which the oscillation is at least that small (Exercise 8.6.6).

Thus we may construct a partition

$$[x_0,x_1],\ [x_1,x_2],\ \ldots,\ [x_{n-1},x_n],$$

of the interval $[a,b]$ consisting of two kinds of closed intervals: (i) the first kind cover all the points of E_1 and have total length smaller than $\varepsilon/(4M+1)$ and (ii) the remaining kind contain no points of E_1 and the oscillation of f on each of these intervals is smaller than $\varepsilon/(2(b-a))$; that is,

$$\omega f([x_{k-1},x_k]) < \varepsilon/(2(b-a)).$$

The sum

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1})$$

splits into two sums depending on the intervals of type (i) or type (ii). The former sum contributes no more than

$$(2M) \times \varepsilon/(4M+1) < \varepsilon/2$$

while the latter sum contributes no more than

$$\varepsilon/(2(b-a)) \times (b-a) = \varepsilon/2.$$

Altogether, then,

$$\sum_{k=1}^{n} \omega f([x_{k-1},x_k])(x_k - x_{k-1}) < \varepsilon$$

and the proof is complete. ■

8.6.4 What Functions Are Riemann Integrable?

✂ *Advanced section. May be omitted.*

Theorem 8.17 exactly characterizes those functions that are Riemann integrable as the class of bounded functions that do not have too many points of discontinuity. We

should recognize immediately that certain types of functions that we are used to working with are also integrable. We express these as corollaries to our theorem. (Recall that step functions were defined in Section 5.2.6.)

Corollary 8.18: *Every step function on an interval $[a,b]$ is Riemann integrable there.*

Proof. A step function is bounded and has only finitely many discontinuities. Thus the set of discontinuities has measure zero. Consequently, the corollary follows from Theorem 8.17. ■

Corollary 8.19: *Every bounded function with only countably many points of discontinuity in an interval $[a,b]$ is Riemann integrable there.*

Proof. The corollary follows directly from Theorem 8.17 since countable sets have measure zero. ■

Corollary 8.20: *Every function monotonic on an interval $[a,b]$ is Riemann integrable there.*

Proof. A monotonic function is bounded and has only countably many discontinuities. Consequently, this corollary follows from the preceding corollary. ■

Corollary 8.21: *If a function f is Riemann integrable on an interval $[a,b]$ then the function $|f|$ is also Riemann integrable on that interval.*

Proof. The corollary follows directly from Theorem 8.17 since if f is Riemann integrable on $[a,b]$ it must be bounded and continuous at every point except a set of measure zero. Exercise 8.6.7 shows that $|f|$ has precisely the same properties. ■

Exercises

8.6.1 Show directly from Theorem 8.16 that the characteristic function of the rationals is not Riemann integrable on any interval.
SEE NOTE 232

8.6.2 Show that the product of two Riemann integrable functions is itself Riemann integrable.

8.6.3 If f is Riemann integrable on an interval and f is never zero, does it follow that $1/f$ is Riemann integrable there? What extra hypothesis could we invoke to make this so?
SEE NOTE 233

8.6.4 If f is Riemann integrable on an interval $[a,b]$ show that for every $\varepsilon > 0$ there are a pair of step functions

$$L(x) \leq f(x) \leq U(x)$$

so that

$$\int_a^b (U(x) - L(x))\,dx < \varepsilon.$$

SEE NOTE 234

8.6.5 Let f be a function on an interval $[a,b]$ with the property that for every $\varepsilon > 0$ there are a pair of step functions $L(x) \leq f(x) \leq U(x)$ so that

$$\int_a^b (U(x) - L(x))\,dx < \varepsilon.$$

Show that f is Riemann integrable.

8.6.6 Suppose that the oscillation $\omega_f(x)$ of a function f is smaller than η at each point x of an interval $[c,d]$. Show that there must be a partition $[x_0,x_1]$, $[x_1,x_2]$, $\ldots$, $[x_{n-1},x_n]$, of $[c,d]$ so that the oscillation

$$\omega f([x_{k-1},x_k]) < \eta$$

on each member of the partition.

SEE NOTE 235

8.6.7 Show that the set of points at which a function F is discontinuous includes all points at which $|F|$ is discontinuous but not conversely. Deduce Corollary 8.21 as a result of this observation from Theorem 8.17.

8.6.8 Deduce Corollary 8.18 directly from Theorem 8.15 rather than from Theorem 8.17.

8.6.9 Deduce Corollary 8.19 directly from Theorem 8.15.

8.6.10 Deduce Corollary 8.20 directly from Theorem 8.15.

8.6.11 Show that the converse of Corollary 8.21 does not hold.

8.7 Properties of the Riemann Integral

✂ The proofs in this section make use of the Lebesgue criterion for integrability. You may skip the proofs and just see how the properties are essentially unchanged from Section 8.3 for Cauchy's original integral.

The Riemann integral is an extension of Cauchy's first integral from continuous functions to a larger class of bounded functions—those that are bounded and continuous except at the points of a small set (a set of measure zero). We have enlarged the class of functions to which the notion of an integral may be applied. Have we lost any of our crucial properties of Section 8.3?

These properties express how we expect integration to behave; it would be distressing to lose any of them. In some cases they remain completely unchanged. In some cases they need to be modified slightly. But our goal was never simply to integrate as many functions as possible; it is to preserve the theory of the integral and to apply that theory sufficiently broadly to handle all necessary applications. If we lose our basic properties we have lost too much. Fortunately the Riemann integral keeps all of the basic properties of the integral of continuous functions. The few differences should be carefully noted. Note especially how some of the properties must be rephrased.

8.22 (Additive Property) *If f is Riemann integrable on both intervals $[a,b]$ and $[b,c]$ then it is Riemann integrable on $[a,c]$ and*

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx.$$

Proof. The proof of the identity need not change from the way we handled it for continuous functions (check this). It is the first assertion in the statement that must be verified. We prove that f is Riemann integrable on $[a,c]$.

By Theorem 8.17 if f is Riemann integrable on both of these intervals it is bounded on both and the set of points of discontinuity in each interval has measure zero. It follows that f is bounded on $[a,c]$. Also, its set of points of discontinuity in $[a,c]$ is the union of the set of points of discontinuity in $[a,b]$ and $[b,c]$ together with (possibly) the point b itself. Thus the set of points of discontinuity in $[a,c]$ is also of measure zero. Consequently, by Theorem 8.17, f is Riemann integrable. ■

8.23 (Linear Property) *If f and g are both Riemann integrable on $[a,b]$, then so too is any linear combination $\alpha f+\beta g$ and*

$$\int_a^b [\alpha f(x)+\beta g(x)]\,dx = \alpha\int_a^b f(x)\,dx+\beta\int_a^b g(x)\,dx.$$

Proof. Again the proof of the identity does not change from the way we handled it for continuous functions (check this). It is the first assertion in the statement that needs to be verified. We must prove that $\alpha f+\beta g$ is Riemann integrable on $[a,b]$.

The points of discontinuity of the function function $\alpha f+\beta g$ are either points of discontinuity of f or else they are points of discontinuity of g. If both functions f and g are Riemann integrable, then they are both bounded and continuous except at the points of a set of measure zero. It follows that $\alpha f+\beta g$ is bounded and continuous except at the points of a set of measure zero. Hence, by Theorem 8.17, $\alpha f+\beta g$ is Riemann integrable. ■

8.24 (Monotone Property) *If f and g are both Riemann integrable on $[a,b]$, then, if $f(x)\le g(x)$ for all $a\le x\le b$,*

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Proof. The proof for continuous functions works equally well here. ■

8.25 (Absolute Property) *If f is Riemann integrable on $[a,b]$, then so too is $|f|$ and*

$$-\int_a^b |f(x)|\,dx \le \int_a^b f(x)\,dx \le \int_a^b |f(x)|\,dx$$

or, equivalently,

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx.$$

Proof. The proof for continuous functions works equally well here because we have already shown, in Corollary 8.21 that if f is Riemann integrable on $[a,b]$, then so too is $|f|$. ■

Fundamental Theorem of Calculus The next two properties, 8.26 and 8.27, are important. They show how the processes of integration and differentiation are inverses of each other. Together they are known as the fundamental theorem of calculus for the Riemann integral. You should note, however, a weakness in this theory. If we compute F' we cannot immediately conclude from 8.27 that $\int_a^b F'(x)\,dx = F(b) - F(a)$. We need first to check that F' is Riemann integrable. This may not always be easy. Worse yet, it may be false, even for bounded derivatives (see the discussion in Section 9.7). It was this failure of the Riemann integral to integrate *all* derivatives that Lebesgue claimed was his motivation to look for a more general theory of integration.

8.26: (Differentiation of the Indefinite Integral) *If f is Riemann integrable on $[a,b]$ then the function*

$$F(x) = \int_a^x f(t)\,dt$$

is continuous on $[a,b]$ and $F'(x) = f(x)$ at each point x at which the function f is continuous.

Proof. Once again, the proof for continuous functions works equally well here. Note, however, that we are no longer trying to prove that $F'(x) = f(x)$ at every point x, only at those points x where f is continuous.

It is left to you to check the proof and verify that it works here, unchanged. ■

8.27 (Integral of a Derivative) *Suppose that the function F is differentiable on $[a,b]$. Provided it is also true that F' is Riemann integrable on $[a,b]$, then*

$$\int_a^b F'(x)\,dx = F(b) - F(a).$$

Proof. Again the proof for continuous functions works here. ■

Exercises

8.7.1 Give a set of conditions under which the integration by substitution formula

$$\int_a^b f(\phi(t))\phi'(t)\,dt = \int_{\phi(a)}^{\phi(b)} f(x)\,dx$$

holds.

SEE NOTE 236

8.7.2 Give a set of conditions under which the integration by parts formula

$$\int_a^b f(t)g'(t)\,dt = f(b)g(b) - f(a)(g(a) - \int_a^b f'(t)g(t)\,dt$$

holds.

8.7.3 Suppose that f is Riemann integrable on $[a,b]$ and define the function

$$F(x) = \int_a^x f(t)\,dt.$$

(a) Show that F satisfies a Lipschitz condition on $[a,b]$; that is, that there exists $M > 0$ such that for every $x, y \in [a,b]$,

$$|F(y) - F(x)| \leq M|y - x|.$$

(b) If x is a point at which f is not continuous is it still possible that $F'(x) = f(x)$?

(c) Is it possible that $F'(x)$ exists but is not equal to $f(x)$?

(d) Is it possible that $F'(x)$ fails to exist?

8.7.4 The function

$$F(x) = \int_0^x \sin(1/t)\,dt$$

has a derivative at every point where the integrand is continuous. Does it also have a derivative at $x = 0$?

8.7.5 Improve Property 8.27 by assuming that F is continuous on $[a,b]$ and allowing that F' exists at all points of $[a,b]$ with finitely many exceptions.

8.7.6 ✂ Do much better than the preceding exercise and improve Property 8.27 by assuming that F is continuous on $[a,b]$ and allowing that F' exists at all points of $[a,b]$ with countably many exceptions.

8.7.7 If f and g are Riemann integrable on an interval $[a,b]$ show that

$$\left(\int_a^b f(x)g(x)\,dx\right)^2 \leq \left(\int_a^b [f(x)]^2\,dx\right)\left(\int_a^b [g(x)]^2\,dx\right).$$

This extends the Cauchy-Schwarz inequality of Exercise 8.3.10.

8.7.8 Show that the integration by parts formula of Exercise 8.3.7 extends to the case where f and g are continuous and f' and g' are Riemann integrable.

8.7.9 ✂ **(More on the Fundamental Theorem of Calculus)** Let f be bounded on $[a,b]$ and continuous a.e. on $[a,b]$. Suppose that F is defined on $[a,b]$ and that $F'(x) = f(x)$ for all x in $[a,b]$ except at the points of some set of measure zero.

(a) Is it necessarily true that $F(x) - F(a) = \int_a^x f(t)\,dt$ for every $x \in [a,b]$?

(b) Same question as in (a) but assume also that F is continuous.

(c) Same question, but this time assume that F is a Lipschitz function. You may assume the nonelementary fact that a Lipschitz function H with $H' = 0$ a.e. must be constant.

(d) Give an example of a Lipschitz function F such that F is differentiable, F' is bounded, but F' is not integrable.

SEE NOTE 237

8.8 The Improper Riemann Integral

✂ *Enrichment section. May be omitted.*

The Riemann integral applies only to bounded functions. What should we mean by the integral

$$\int_0^1 \frac{dx}{\sqrt{x}}?$$

Since the integrand is unbounded on $[0,1]$, it is not Riemann integrable even though the integrand is continuous at all but one point. There is not much else for us to do but to back track by several decades and return to Cauchy's second method; namely, we compute

$$\lim_{\delta\to 0+} F(\delta) = \lim_{\delta\to 0+} \int_{\delta}^{1} \frac{dx}{\sqrt{x}} = 2.$$

What we should probably do now is to create a new hybrid integral by combining the Riemann integral with Cauchy's second method. This is often called the *improper Riemann integral.* As before we give a definition that considers only one point of unboundedness (at the left endpoint of the interval) with the understanding that the ideas can be applied to any finite number of such points.

Definition 8.28: Let f be a function on an interval $(a,b]$ that is Riemann integrable on $[a+\delta,b]$ and that is unbounded in the interval $(a,a+\delta)$ for every $0<\delta<b-a$. Then we define

$$\int_a^b f(x)\,dx$$

to be

$$\lim_{\delta\to 0+} \int_{a+\delta}^{b} f(x)\,dx$$

if this limit exists, and in this case the integral is said to be *convergent*. If both integrals

$$\int_a^b f(x)\,dx \text{ and } \int_a^b |f(x)|\,dx$$

converge the integral is said to be *absolutely convergent*.

In the same way we also extend the Riemann integral from bounded intervals to unbounded ones. How should we interpret the integral

$$\int_1^{\infty} \frac{dx}{x^2}?$$

This cannot exist as a Riemann integral since the definition is clearly restricted to finite intervals and would not allow any easy interpretation for infinite intervals. As before we use Cauchy's second method to obtain

$$\int_1^{\infty} \frac{dx}{x^2} = \lim_{X\to\infty} \int_1^{X} \frac{dx}{x^2} = \lim_{X\to\infty} \left(1-\frac{1}{X}\right) = 1.$$

We give a formal definition valid just for an infinite interval of the form $[a,\infty)$. The case $(-\infty,b]$ is similar. The case $(-\infty,+\infty)$ is best split up into the sum of two integrals, from $(-\infty,a]$ and $[a,\infty)$, each of which can be handled in this fashion.

Definition 8.29: Let f be a function on an interval $[a,\infty)$ that is Riemann integrable on every interval $[a,b]$ for $a<b<\infty$. Then we define

$$\int_a^{\infty} f(x)\,dx$$

to be

$$\lim_{X\to\infty}\int_a^X f(x)\,dx$$

if this limit exists, and in this case the integral is said to be *convergent*. If both integrals

$$\int_a^\infty f(x)\,dx \text{ and } \int_a^\infty |f(x)|\,dx$$

converge the integral is said to be *absolutely convergent*.

Both of these definitions extend the Riemann integral to a more general concept. Note that in any applications using an improper Riemann integral of either type, we are obliged to announce whether the integral is convergent or divergent, and frequently whether it is absolutely or nonabsolutely convergent.

It might seem that this theory would be important to master and represents the final word on the subject of integration. By the end of the nineteenth century it had become increasingly clear that this theory of the Riemann integral itself was completely inadequate to handle the bounded functions that were arising in many applications. The extra step here, using Cauchy's second method, designed to handle unbounded functions, also proved far too restrictive. The modern theory of integration was developed in the first decades of the twentieth century. The methods are different and even the language needs many changes.

Thus, the material in these last few sections has largely a historical interest. Some mathematicians claim it has only that, others that learning this material is a good preparation for learning the more advanced material.

Exercises

8.8.1 For what values of p, q are the integrals

$$\int_0^1 \frac{\sin x}{x^p}\,dx \text{ and } \int_0^1 \frac{(\sin x)^q}{x}\,dx$$

ordinary Riemann integrals, convergent improper Riemann integrals, or divergent improper Riemann integrals?

8.9 More on the Fundamental Theorem of Calculus

✂ *Enrichment section. May be omitted.*

The Riemann integral does not integrate all bounded derivatives and so the fundamental theorem of calculus for this integral assumes the awkward form

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

provided F is differentiable on $[a,b]$ and *the derivative F' is Riemann integrable there.*

The emphasized phrase is unfortunate. It means we have a limited theory and it also means that, in practice, we must always check to be sure that a derivative F' is integrable before proceeding to integrate it. In Section 9.7 we shall show how to construct a function F that is everywhere differentiable on an interval and whose derivative F' is bounded but not itself Riemann integrable on any subinterval.

Let us take another look at the integrability of derivatives to see if we can discover what goes wrong. We take a completely naive approach and start with the definition of the derivative itself. If $F' = f$ everywhere, then, at each point ξ and for every $\varepsilon > 0$, there is a $\delta > 0$ so that

$$|F(x'') - F(x') - f(\xi)(x'' - x')| < \varepsilon(x'' - x') \tag{6}$$

for $x' \le \xi \le x''$ and $0 < x'' - x' < \delta$.

A careless student might argue that one can recover $F(b) - F(a)$ as a limit of Riemann sums for f as follows. Let

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

be a partition of $[a,b]$, and let $\xi_i \in [x_{i-1}, x_i]$. Then

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_{i-1}) - F(x_i)) = \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}) + R$$

where

$$R = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1}) - f(\xi_i)(x_i - x_{i-1})).$$

Thus $F(b) - F(a)$ has been given as a Riemann sum for f plus some error term R. But it appears now that, if the partition is finer than the number δ so that (6) may be used, we have

$$\begin{aligned} |R| &\le \sum_{i=1}^{n} \left| F(x_i) - F(x_{i-1}) - f(\xi_i)(x_i - x_{i-1}) \right| \\ &< \sum_{i=1}^{n} \varepsilon(x_i - x_{i-1}) = \varepsilon(b-a). \end{aligned}$$

Evidently, then, *if there are no mistakes here* it follows that f is Riemann integrable and that $\int_a^b f(t)\,dt = F(b) - F(a)$.

This is false, as we have mentioned, and we leave if for you in Exercise 8.9.1 to spot the error. But, instead of abandoning the argument, we can change the definition of the Riemann integral to allow this argument to work. The definition changes to look like this.

Definition 8.30: A function f is *generalized Riemann integrable* on $[a,b]$ with value I if for every ε there is a positive function δ on $[a,b]$ so that

$$\left| \sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}) - I \right| < \varepsilon$$

whenever

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$

is a partition of $[a,b]$ with $\xi_i \in [x_{i-1}, x_i]$ and $0 < x_i - x_{i-1} < \delta(\xi_i)$.

The integral $\int_a^b f(x)\,dx$ is taken as this number I that exists. It is easy to check that if f is Riemann integrable it is also generalized Riemann integrable and the integrals have the same value. Thus this new integral is an extension of the old one. To justify

the definition requires knowing that such partitions actually exist for any such positive function δ; this is supplied by the Cousin theorem (Lemma 4.26).

This defines a Riemann-type integral that includes the usual Riemann integral and integrates *all* derivatives. The generalized Riemann integral was discovered in the 1950s, independently, by R. Henstock and J. Kurzweil, and these ideas have led to a number of other integration theories that exploit the geometry of the underlying space in the same way that this integral exploits the geometry of derivatives on the real line.

We shall not carry these ideas any further but refer you to the monographs of Pfeffer[1] or Gordon.[2]

Exercises

8.9.1 Spot the error in the careless student argument given in the text.

SEE NOTE 238

8.9.2 Develop the elementary properties of the generalized Riemann integral directly from its definition.

SEE NOTE 239

8.9.3 Show directly from the definition that the characteristic function of the rationals is not Riemann integrable but is generalized Riemann integrable on any interval and that $\int_0^1 f(x)\,dx = 0$.

8.9.4 Show that the generalized Riemann integral is closed under the extension procedure of Cauchy from Section 8.4.

8.10 Challenging Problems for Chapter 8

8.10.1 Let $m(f,\pi)$ and $M(f,\pi)$ denote the upper and lower sums over a partition π for a bounded function f. Define the upper and lower integrals as

$$\overline{\int_a^b} f(x)\,dx = \inf M(f,\pi) \quad \text{and} \quad \underline{\int_a^b} f(x)\,dx = \inf m(f,\pi)$$

where the inf and sup are taken over all possible partitions π of the interval $[a,b]$. We say f is *Darboux integrable* if the upper and lower integrals are equal.

(a) Show that $\underline{\int_a^b} f(x)\,dx \le \overline{\int_a^b} f(x)\,dx$.

(b) Show that every Riemann integrable function is Darboux integrable.

(c) Show that every Darboux integrable function is Riemann integrable.

(d) Show that if f is Riemann integrable, then

$$\overline{\int_a^b} f(x)\,dx = \underline{\int_a^b} f(x)\,dx = \int_a^b f(x)\,dx.$$

[1] W. F. Pfeffer, *The Riemann Approach to Integration: Local Geometric Theory.* Cambridge University Press (1993).

[2] R. A. Gordon, *The Integrals of Lebesgue, Denjoy, Perron and Henstock.* Grad. Studies in Math., 4, Amer. Math. Soc. (1994).

(e) Show that

$$\overline{\int_a^b}(f(x)+g(x))\,dx \le \overline{\int_a^b} f(x)\,dx + \overline{\int_a^b} g(x)\,dx$$

with strict inequality possible.

SEE NOTE 240

8.10.2 Let $f:[0,1]\to\mathbb{R}$ be a differentiable function such that $|f'(x)|\le M$ for all $x\in(0,1)$. Show that

$$\left|\int_0^1 f(x)\,dx - \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right)\right| \le \frac{M}{n}.$$

SEE NOTE 241

Notes

[226]Exercise 8.2.1. You will need to find a formula for

$$\sum_{k=1}^{n} k^3.$$

[227]Exercise 8.2.9. Be sure, first, to check that these associated points are legitimate. Show that each of these sums has the same value (think of telescoping sums!). What, then, would be the limit of the Riemann sums?

[228]Exercise 8.3.10. This is called the *Cauchy-Schwarz inequality* and is the analog for integrals of that inequality in Exercise 3.5.13. It can be proved the same way and does not involve any deep properties of integrals.

[229]Exercise 8.4.5. It would converge for all continuous functions.

[230]Exercise 8.5.2. Define

$$\int_{-\infty}^{\infty} f(x)\,dx$$

to be the sum of

$$\int_{-\infty}^{a} f(x)\,dx$$

and

$$\int_{a}^{\infty} f(x)\,dx.$$

Be sure to prove that this definition would not depend on the choice of a.

[231]Exercise 8.5.10. Compare with Exercise 3.4.26. Note, too, that it may seem to require special handling at the left-hand endpoint but it does not.

[232]Exercise 8.6.1. Note that this function is discontinuous everywhere and that

$$\omega f([c,d]) = 1$$

for every interval $[c,d]$.

[233]Exercise 8.6.3. The answer is no. It would be true if $|f|>c>0$ everywhere. Equivalently, it is true if $1/f$ is bounded.

[234]Exercise 8.6.4. Step functions were defined in Section 5.2.6. If you sketch a picture of what the approximating sums look like, the step functions needed should be apparent.

[235]Exercise 8.6.6. The fact that the oscillation of a function f is smaller than η at each point of an interval $[c,d]$ is a local condition. Express it by using a $\delta(x)$ at each point. Now use a compactness argument (e.g., Heine-Borel) to get a uniform size that works.

[236]Exercise 8.7.1. Make ϕ' integrable and f continuous at each point $\phi(t)$ for $t \in [a,b]$.

[237]Exercise 8.7.9. For (a): What if F is discontinuous? For (b): Consider the Cantor function (Section 6.5.3). For (d): This is not easy! We will discuss this in Section 9.7.

[238]Exercise 8.9.1. The error is that the choice of δ depends on the point ξ considered and so is not a constant. This is an error you have doubtless made in other contexts: A local condition that holds for *each* point x is misinterpreted as holding uniformly for *all* x.

[239]Exercise 8.9.2. Consider the integral of a sum $f+g$, the integral formula $\int_a^b + \int_b^c = \int_a^c$, etc.x

[240]Exercise 8.10.1. This exercise develops the theory of the *Darboux integral*, which is equivalent to Riemann's integral but defined using infs and sups of "Darboux sums" rather than limits of Riemann sums. In preparation Exercise 8.2.17 should be consulted.

[241]Exercise 8.10.2. This is from the 1947 Putnam Mathematical Competition.

Chapter 9

SEQUENCES AND SERIES OF FUNCTIONS

✂ If the material on series in Chapter 3 was omitted in a first reading, then Sections 3.4, 3.5, and parts of 3.6 should be studied before attempting this chapter.

9.1 Introduction

We have seen that a function f that is the sum of two or more functions will share certain desirable properties with those functions. For example, our study of continuity, differentiation, and integration allows us to state that if

$$f = f_1 + f_2 + \cdots + f_n$$

on an interval $I = [a,b]$, then

(1) If $f_1, f_2, \ldots, f_n$ are continuous on I, so is f.

(2) If $f_1, f_2, \ldots, f_n$ are differentiable on I, so is f, and

$$f' = f_1' + f_2' + \cdots + f_n'.$$

(3) If $f_1, f_2, \ldots, f_n$ are integrable on I, so is f, and

$$\int_a^b f(x)\,dx = \int_a^b f_1(x)\,dx + \int_a^b f_2(x)\,dx + \cdots + \int_a^b f_n(x)\,dx.$$

It is natural to ask whether the corresponding results hold when f is the sum of an *infinite* series of functions,

$$f = \sum_{k=0}^{\infty} f_k.$$

If each term of the series is continuous, is the sum function also continuous? Can the derivative be obtained by summing the derivatives? Can the integral be obtained by summing the integrals? We study such questions in this chapter.

These problems are of considerable practical importance. For example, if we are allowed to take limits, integrate, and differentiate freely, then the computations in the following example would all be valid.

Example 9.1: From the formula for the sum of a geometric series we know that

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots \tag{1}$$

on the interval $(-1,1)$. Differentiation of both sides of (1) leads immediately to

$$\frac{-1}{(1+x)^2} = -1 + 2x - 3x^2 + 4x^3 - 5x^4 + \dots.$$

Repeated differentiation would give formulas for $(1+x)^{-n}$ for all positive integers n.

On the other hand, integration of both sides of (1) from 0 to t leads immediately to

$$\ln(1+t) = t - \frac{1}{2}t^2 + \frac{1}{3}t^3 - \frac{1}{4}t^4 + \frac{1}{5}t^5 - \dots.$$

Taking limits as $t \to 1$ in the latter yields the intriguing formula for the sum of the alternating harmonic series:

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

◀

The conclusions in the example are all true and useful. But have we used illegitimate means to find them? If we use such methods freely might we find situations where our conclusions are wrong?

We first formulate our questions in the language of sequences of functions (rather than series). We do this in Section 9.2, where we see that the answer to our questions is "not necessarily." Then in Sections 9.3–9.6 we see that if we require a bit more of convergence, the answer to each of our questions is "yes."

9.2 Pointwise Limits

Suppose $f_1, f_2, f_3, \dots$ is a sequence of functions, each of which is defined on a common domain D. What should we mean by the sum

$$f = \sum_{k=0}^{\infty} f_k$$

or by the limit

$$f = \lim_{n\to\infty} f_n?$$

For sequence limits it would seem natural to require that the values of the functions $f_n(x)$ should converge to the values of the function $f(x)$.

Definition 9.2: Let $\{f_n\}$ be a sequence of functions defined on a common domain D. If $\lim_{n\to\infty} f_n(x)$ exists (as a real number) for all $x \in D$, we say that the sequence $\{f_n\}$ *converges pointwise* on D. This limit defines a function f on D by the equation

$$f(x) = \lim_{n} f_n(x).$$

We write $\lim_n f_n = f$ or $f_n \to f$.

For the infinite sum, the simplest idea is to extend the definition of finite sum using our familiar interpretation of convergence of an infinite series of numbers as a limit of the sequence of partial sums.

Definition 9.3: For each x in D and $n \in \mathbb{N}$ let

$$S_n(x) = f_1(x) + \cdots + f_n(x).$$

If $\lim_{n\to\infty} S_n(x)$ exists (as a real number), we say the series $\sum_{k=1}^{\infty} f_k$ *converges at* x and we write

$$\sum_{k=1}^{\infty} f_k(x)$$

for $\lim_{n\to\infty} S_n(x)$. If the series converges for all $x \in D$, we say the series *converges pointwise* on D to the function f defined by

$$f(x) = \sum_{k=1}^{\infty} f_k(x) \quad (= \lim_{n\to\infty} \sum_{k=1}^{n} f_k(x)).$$

We would like such infinite sums of functions to behave like finite sums of functions (as our three questions in Section 9.1 suggest): If $f = \sum_1^{\infty} f_k$ on an interval $I = [a,b]$, is it true that

(1) If f_k is continuous on I for all $k \in \mathbb{N}$, then so is f?

(2) If f_k is differentiable on I for all $k \in \mathbb{N}$, then so is f and

$$f'(x) = \sum_{k=1}^{\infty} f_k'(x)?$$

(3) If f_k is integrable on I for all $k \in \mathbb{N}$, then so is f, and

$$\int_a^b f(x)\,dx = \sum_{k=1}^{\infty} \int_a^b f_k(x)\,dx?$$

In the special case that D is an interval $I = [a,b]$ our questions then become the following: Is it true that

1. If f_n is continuous on I for all n, then is f continuous on I?
2. If f_n is differentiable on I for all n, then is f differentiable on I and, if so, does $f' = \lim_n f_n'$?
3. If f_n is integrable on I for all n, then is f integrable on I and, if so, does $\int_a^b f(x)\,dx = \lim_n \int_a^b f_n(x)\,dx$?

These questions have *negative* answers in general, as the three examples that follow show.

Example 9.4: (A discontinuous limit of continuous functions) For each $n \in \mathbb{N}$ and $x \in [0,1]$, let $f_n(x) = x^n$. Each of the functions is continuous on $[0,1]$. Notice, however, that for each $x \in (0,1)$, $\lim_n f_n(x) = 0$ and yet $\lim_n f_n(1) = 1$. This is easy to see, but it is instructive to check the details since we can use them later to see what is going wrong in this example. At the right-hand endpoint it is clear that, for $x = 1$, $\lim_n f_n(x) = 1$. For $0 < x_0 < 1$ and $\varepsilon > 0$, let $N \geq \ln\varepsilon / \ln x_0$. Then $(x_0)^N \leq \varepsilon$, so for $n \geq N$

$$|f_n(x_0) - 0| = (x_0)^n < (x_0)^N \leq \varepsilon.$$

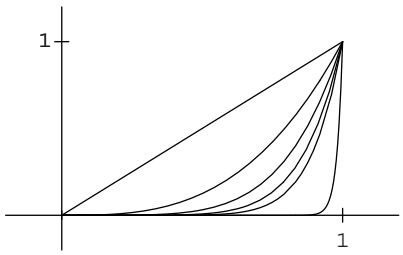


Figure 9.1. Graphs of x^n on $[0,1]$ for $n = 1, 3, 5, 7, 9$, and 50.

Thus

$$f(x) = \lim_n f_n(x) = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1, \end{cases}$$

so the pointwise limit f of the sequence of continuous functions $\{f_n\}$ is discontinuous at $x = 1$. (Figure 9.1 shows the graphs of several of the functions in the sequence.) ◀

Example 9.5: (The derivative of the limit is not the limit of the derivative.) Let $f_n(x) = x^n/n$. Then $f_n \to 0$ on $[0,1]$. Now $f_n'(x) = x^{n-1}$, so by the previous example, Example 9.4,

$$\lim_n f_n'(x) = \lim_n x^{n-1} = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1, \end{cases}$$

while the derivative of the limit function, $f \equiv 0$, equals zero on $[0,1]$. Thus

$$\lim_{n\to\infty} \frac{d}{dx}(f_n(x)) \neq \frac{d}{dx}\left(\lim_{n\to\infty} f_n(x)\right)$$

at $x = 1$. ◀

Example 9.6: (The integral of the limit is not the limit of the integrals.) In this example we consider a sequence of continuous functions, each of which has the same integral over the domain. For each $n \in \mathbb{N}$ let f_n be defined on $[0,1]$ as follows: $f_n(0) = 0$, $f_n(1/(2n)) = 2n$, $f_n(1/n) = 0$, f_n is linear on $[0, 1/(2n)]$ and on $[1/(2n), 1/n]$, and $f_n = 0$ on $[1/n, 1]$. (See Figure 9.2.)

It is easy to verify that $f_n \to 0$ on $[0,1]$. Now, for each $n \in \mathbb{N}$,

$$\int_0^1 f_n(x)\, dx = 1.$$

But

$$\int_0^1 (\lim_n f_n(x))\, dx = \int_0^1 0\, dx = 0.$$

Thus

$$\lim_n \int_0^1 f_n(x)\, dx \neq \int_0^1 \lim_n f_n(x)\, dx$$

so that the limit of the integrals is not the integral of the limit. ◀

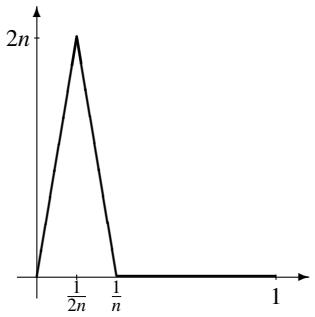


Figure 9.2. Graph of $f_n(x)$ on $[0,1]$ in Example 9.6.

These examples show that the answer to each of our three questions is negative, in general. We present some additional examples that illustrate similar phenomena in the exercises.

We shall see in the next few sections that by replacing pointwise convergence in appropriate places with a stronger form of convergence, the answers to our questions become affirmative. The form of convergence in question is called *uniform convergence*.

Interchange of Limit Operations Before turning to uniform convergence, let us first try to get an insight into a difficulty we must overcome if we wish affirmative answers to our questions. If f_n is a sequence of continuous functions converging to a function f, must f be continuous? Continuity of f at a point x_0 would mean that

$$\lim_{x\to x_0} f(x) = f(x_0)$$

and this would require that

$$\lim_{x\to x_0}(\lim_{n\to\infty} f_n(x)) = \lim_{n\to\infty} f_n(x_0) = \lim_{n\to\infty}(\lim_{x\to x_0} f_n(x)).$$

Apparently, to verify the continuity of f at x_0 we need to use two limit operations and be assured that the order of passing to the limits is immaterial.

You will remember situations in which two limit operations are involved and the order of taking the limit does not affect the result. For example, in elementary calculus one finds conditions under which the value of a double integral can be obtained by iterating "single integrals" in either order. By way of contrast, we present an example in the setting of double sequences in which the order of taking limits *is* important.

Example 9.7: In this example we illustrate that an interchange of limit operations may not give a correct result. Let

$$S_{mn} = \begin{cases} 0, & \text{if } m \le n \\ 1, & \text{if } m > n. \end{cases}$$

Viewed as a matrix,

$$[S_{mn}] = \begin{bmatrix} 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & \cdots \\ 1 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

where we are placing the entry S_{mn} in the mth row and nth column. For each row m, we have $\lim_{n\to\infty} S_{mn} = 0$, so

$$\lim_{m\to\infty} \left(\lim_{n\to\infty} S_{mn} \right) = 0.$$

On the other hand, for each column n, $\lim_{m\to\infty} S_{mn} = 1$, so

$$\lim_{n\to\infty} \left(\lim_{m\to\infty} S_{mn} \right) = 1.$$

◀

Exercises

9.2.1 Examine the pointwise limiting behavior of the sequence of functions

$$f_n(x) = \frac{x^n}{1+x^n}.$$

9.2.2 Show that the logarithm function can be expressed as the pointwise limit of a sequence of "simpler" functions,

$$\ln x = \lim_{n\to\infty} n\left(\sqrt[n]{x} - 1\right)$$

for every point in its domain. If the answer to our three questions for this particular limit is affirmative, what can you say about the continuity of the logarithm function? What would be its derivative? What would be $\int_1^2 \ln x\,dx$?

9.2.3 Let $x_1, x_2, \ldots$ be an enumeration of $\mathbb{Q}$, let

$$f_n(x) = \begin{cases} 1, & \text{if } x \in \{x_1, \ldots, x_n\} \\ 0, & \text{otherwise,} \end{cases}$$

and let

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{otherwise.} \end{cases}$$

Show that $f_n \to f$ pointwise on $[0,1]$, but $\int_0^1 f_n(x)\,dx = 0$ for all $n \in \mathbb{N}$, while f is not integrable on $[0,1]$.

9.2.4 Let $f_n(x) = \sin nx/\sqrt{n}$. Show that $\lim_n f_n = 0$ but $\lim_n f_n'(0) = \infty$.

9.2.5 Each of Examples 9.4, 9.5 and 9.6 can be interpreted as a statement that the order of taking the limit operation does matter. Verify this.

9.2.6 Refer to Example 9.7. What should we mean by the statement that a "double sequence" $\{t_{mn}\}$ converges; that is, that

$$\lim_{m\to\infty, n\to\infty} t_{mn}$$

exists? Does the double sequence $\{S_{mn}\}$ of Example 9.7 converge?

9.2.7 Let $f_n \to f$ pointwise at every point in the interval $[a,b]$. We have seen that even if each f_n is continuous it does not follow that f is continuous. Which of the following statements are true?

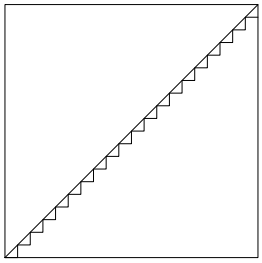


Figure 9.3. Construction in Exercise 9.2.8.

(a) If each f_n is increasing on $[a,b]$, then so is f.

(b) If each f_n is nondecreasing on $[a,b]$, then so is f.

(c) If each f_n is bounded on $[a,b]$, then so is f.

(d) If each f_n is everywhere discontinuous on $[a,b]$, then so is f.

(e) If each f_n is constant on $[a,b]$, then so is f.

(f) If each f_n is positive on $[a,b]$, then so is f.

(g) If each f_n is linear on $[a,b]$, then so is f.

(h) If each f_n is convex on $[a,b]$, then so is f.

SEE NOTE 242

9.2.8 A careless student[1] once argued as follows: "It seems to me that one can construct a curve without a tangent in a very elementary way. We divide the diagonal of a square into n equal parts and construct on each subdivision as base a right isosceles triangle. In this way we get a kind of delicate little saw. Now I put $n = \infty$. The saw becomes a continuous curve that is infinitesimally different from the diagonal. But it is perfectly clear that its tangent is alternately parallel now to the x-axis, now to the y-axis." What is the error? (Figure 9.3 illustrates the construction.)

SEE NOTE 243

9.2.9 As yet another illustration that some properties are not preserved in the limit, compute the length of the curves in Exercise 9.2.8 (Fig. 9.3) and compare with the length of the limiting curve.

9.2.10 ✂ If $f_n \to f$ pointwise at every real number, then prove that

$$\{x : f(x) > \alpha\} = \bigcup_{m=1}^{\infty} \bigcup_{r=1}^{\infty} \bigcap_{n=r}^{\infty} \{x : f_n(x) \geq \alpha + 1/m\}.$$

9.2.11 Let $\{f_n\}$ be a sequence of real functions. Show that the set E of points of convergence of the sequence can be written in the form

$$E = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \bigcap_{m=N}^{\infty} \left\{x : |f_n(x) - f_m(x)| \leq \tfrac{1}{k}\right\}.$$

[1] In this case the "careless student" was the great Russian analyst N. N. Luzin (1883–1950), who recounted in a letter [reproduced in *Amer. Math. Monthly*, 107, (2000), pp. 64–82] how he offered this argument to his professor after a lecture on the Weierstrass continuous nowhere differentiable function.

9.3 Uniform Limits

Pointwise limits do not allow the interchange of limit operations. In many situations, uniform limits will. To see how the definition of a uniform limit needs to be formulated, let us return to the sequence of Example 9.4. That sequence illustrated the fact that a pointwise limit of continuous functions need not be continuous. The difficulty there was that

$$\lim_{x\to 1-}\left(\lim_{n\to\infty} f_n(x)\right) \neq \lim_{n\to\infty}\left(\lim_{x\to 1-} f_n(x)\right).$$

A closer look at the limits involved here shows what went wrong and suggests what we need to look for in order to allow an interchange of limits.

Example 9.8: Consider again the sequence $\{f_n\}$ of functions $f_n(x) = x^n$. We saw that $f_n \to 0$ pointwise on $[0,1)$, and that for every fixed $x_0 \in (0,1)$ and $\varepsilon > 0$,

$$|x_0|^n < \varepsilon \text{ if and only if } n > \ln\varepsilon/\ln x_0.$$

Now fix ε but let the point x_0 vary. Observe that, when x_0 is relatively small in comparison with ε, the number $\ln x_0$ is large in absolute value compared with $\ln\varepsilon$, so relatively small values of n suffice for the inequality $|x_0|^n < \varepsilon$. On the other hand, when x_0 is near 1, $\ln x_0$ is small in absolute value, so $\ln\varepsilon/\ln x_0$ will be large. In fact,

$$\lim_{x_0\to 1-} \frac{\ln\varepsilon}{\ln x_0} = \infty. \tag{2}$$

The following table illustrates how large n must be before $|x_0^n| < \varepsilon$ for $\varepsilon = .1$.

x_0	n
.1	2
.5	4
.9	22
.99	230
.999	2,302
.9999	23,025

Note that for $\varepsilon = .1$, there is no single value of N such that $|x_0|^n < \varepsilon$ for every value of $x_0 \in (0,1)$ and $n > N$. (Figure 9.4 illustrates this.) ◀

Some nineteenth-century mathematicians would have described the varying rates of convergence in the example by saying that "the sequence $\{x^n\}$ converges *infinitely slowly* on $(0,1)$." Today we would say that this sequence, which does converge pointwise, does *not* converge uniformly. Our definition is formulated precisely to avoid this possibility of infinitely slow convergence.

Definition 9.9: Let $\{f_n\}$ be a sequence of functions defined on a common domain D. We say that $\{f_n\}$ *converges uniformly* to a function f on D if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \varepsilon \text{ for all } n \geq N \text{ and } x \in D.$$

We write

$$f_n \to f \text{ [unif] on } D \text{ or } \lim_n f_n = f \text{ [unif] on } D$$

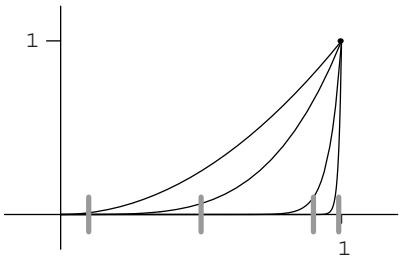


Figure 9.4. The sequence $\{x^n\}$ converges infinitely slowly on $[0,1]$. The functions $y = x^n$ are shown with $n = 2$, 4, 22, and 100, with $x_0 = .1$, .5, .9, and .99, and with $\varepsilon = .1$.

to indicate that the sequence $\{f_n\}$ converges uniformly to f on D. If the domain D is understood from the context, we may delete explicit reference to D and write

$$f_n \to f \text{ [unif] or } \lim_n f_n = f \text{ [unif].}$$

Uniform convergence plays an important role in many parts of analysis. In particular, it figures in questions involving the interchanging of limit processes such as those we discussed in Section 9.2. This was not apparent to mathematicians in the early part of the nineteenth century. As late as 1823, Cauchy believed a convergent series of continuous functions could be integrated term by term. Similarly, Cauchy believed that a convergent series of continuous functions has a continuous sum. Abel provided a counterexample in 1826. It may have been Weierstrass who first recognized the importance of uniform convergence in the middle of the nineteenth century.[2]

Example 9.10: Let $f_n(x) = x^n$, $D = [0,\eta]$, $0 < \eta < 1$. We observed that the sequence $\{f_n\}$ converges pointwise, but not uniformly, on $(0,1)$ (or on $[0,1]$). We realized that the difficulty arises from the fact that the convergence near 1 is very "slow." But for any fixed η with $0 < \eta < 1$, the convergence *is* uniform on $[0,\eta]$.

To see this, observe that for $0 \le x_0 < \eta$, $0 \le (x_0)^n < \eta^n$. Let $\varepsilon > 0$. Since $\lim_n \eta^n = 0$, there exists N such that if $n \ge N$, then $0 < \eta^n < \varepsilon$. Thus, if $n \ge N$, we have

$$0 \le x_0^n < \eta^n < \varepsilon,$$

so the same N that works for $x = \eta$, also works for all $x \in [0,\eta)$. ◀

Suppose that $f_n \to f$ on $[0,1]$. It follows easily from the definition that the convergence is uniform on any *finite* subset D of $[0,1]$ (Exercise 9.3.3). Thus given any $\varepsilon > 0$ and any finite set $x_1, x_2, \ldots, x_m$ in [0,1], we can find $n \in \mathbb{N}$ such that

$$|f_n(x_i) - f(x_i)| < \varepsilon$$

for all $n \ge N$ and all $i = 1, 2, \ldots, m$. (Figure 9.5 illustrates this.)

[2]More on the history of uniform convergence can be found in Thomas Hawkins' interesting historical book *Lebesgue's Theory of Integration: Its Origins and Development*, Univ. of Wisconsin Press (Madison, 1970)

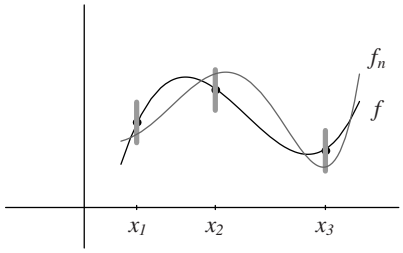


Figure 9.5. Uniform convergence on the finite set $\{x_1, x_2, x_3\}$.

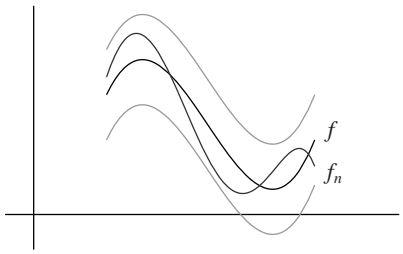


Figure 9.6. Uniform convergence on the whole interval.

The vertical line segments over the points $x_1, \ldots, x_m$ are centered on the graph of f and are of length 2ε. In simple geometric language, we can go sufficiently far out in the sequence to guarantee that the graphs of all the functions f_n intersect all of these finitely many vertical segments.

In contrast, uniform convergence on $[0,1]$ requires that we can go sufficiently far out in the sequence to guarantee that the graphs of the functions go through such vertical segments at *all* points of $[0,1]$; that is, that the graph of f_n for n sufficiently large lies in the "ε-band" centered on the graph of f. (See Figure 9.6.)

9.3.1 The Cauchy Criterion

Suppose now that we are given a sequence of functions $\{f_n\}$ on an interval I, and we wish to know whether it converges uniformly to some function on I. We are not told what that limit function might be. The problem is similar to one we faced for a sequence of *numbers* $\{a_n\}$ in our study of sequences. There we saw that $\{a_n\}$ converges if and only if it is a Cauchy sequence. We can formulate a similar criterion for uniform convergence of a sequence of functions.

Definition 9.11: Let $\{f_n\}$ be a sequence of functions defined on a set D. The sequence is said to be *uniformly Cauchy* on D if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that if $n \geq N$ and $m \geq N$, then $|f_m(x) - f_n(x)| < \varepsilon$ for all $x \in D$.

Theorem 9.12 (Cauchy Criterion) *Let $\{f_n\}$ be a sequence of functions defined on a set D. Then there exists a function f defined on D such that $f_n \to f$ uniformly on D if*

and only if $\{f_n\}$ is uniformly Cauchy.

Proof. We leave the proof of Theorem 9.12 as Exercise 9.3.15. ■

Example 9.13: In Example 9.10 we showed that the sequence $f_n(x) = x^n$ converges uniformly on any interval $[0,\eta]$, for $0 < \eta < 1$. Let us prove this again, but using the Cauchy criterion.

Fix $n \geq m$ and compute

$$\sup_{x\in[0,\eta]} |x^n - x^m| \leq \eta^m. \tag{3}$$

Let $\varepsilon > 0$ and choose an integer N so that $\eta^N < \varepsilon$. Equivalently we require that $N > \ln\varepsilon/\ln\eta$. Then it follows from (3) for all $n \geq m \geq N$ and all $x \in [0,\eta]$ that

$$|x^n - x^m| \leq \eta^m < \varepsilon.$$

We conclude, by the Cauchy criterion, that the sequence $f_n(x) = x^n$ converges uniformly on any interval $[0,\eta]$, for $0 < \eta < 1$. Here there was no computational advantage over the argument in Example 9.10. Frequently, though, we do not know the limit function and *must* use the Cauchy criterion rather than the definition. ◀

Cauchy Criterion for Series The Cauchy criterion can be expressed for uniformly convergent series too. We say that a series $\sum_1^\infty f_k$ converges *uniformly* to the function f on D if the sequence $\{S_n\} = \{\sum_{k=1}^n f_k\}$ of partial sums converges uniformly to f on D.

Theorem 9.14 (Cauchy Criterion) *Let $\{f_k\}$ be a sequence of functions defined on a set D. Then the series $\sum_1^\infty f_k$ converges uniformly to some function f on D if and only if for every $\varepsilon > 0$ there is an integer N so that*

$$\left|\sum_{j=m}^{n} f_j(x)\right| < \varepsilon$$

for all $n \geq m \geq N$ and all $x \in D$.

Proof. This follows immediately from Theorem 9.12. ■

Example 9.15: Let us show that the series

$$1 + x + x^2 + x^3 + x^4 + \ldots$$

converges uniformly on any interval $[0,\eta]$, for $0 < \eta < 1$. Our computations could be based on the fact that the sum of this series is known to us; it is $(1-x)^{-1}$. We could prove the uniform convergence directly from the definition. Instead let us use the Cauchy criterion.

Fix $n \geq m$ and compute

$$\sup_{x\in[0,\eta]} \left|\sum_{j=m}^{n} x^j\right| \leq \sup_{x\in[0,\eta]} \left|\frac{x^m}{1-x}\right| \leq \frac{\eta^m}{1-\eta}. \tag{4}$$

Let $\varepsilon > 0$. Since

$$\eta^m(1-\eta)^{-1} \to 0$$

as $m \to \infty$ we may choose an integer N so that

$$\eta^N (1-\eta)^{-1} < \varepsilon.$$

Then it follows from (4) for all $n \geq m \geq N$ and all $x \in [0,\eta]$ that

$$\left|x^m + x^{m+1} + \cdots + x^n\right| \leq \frac{\eta^m}{1-\eta} < \varepsilon.$$

It follows now, by the Cauchy criterion, that the series converges uniformly on any interval $[0,\eta]$, for $0 < \eta < 1$. Observe, however, that the series does not converge uniformly on $(-1,1)$, though it does converge pointwise there. (See Exercise 9.3.16.) ◀

9.3.2 Weierstrass M-Test

It is not always easy to determine whether a sequence of functions is uniformly convergent. In the settings of *series* of functions, a certain simple test is often useful. This will certainly become one of the most frequently used tools in your study of uniform convergence.

Theorem 9.16 (M-Test) *Let $\{f_k\}$ be a sequence of functions defined on a set D and let $\{M_k\}$ be a sequence of positive constants. If*

$$\sum_0^\infty M_k < \infty$$

and if

$$|f_k(x)| \leq M_k$$

for each $x \in D$ and $k = 0,1,2,\ldots$, then the series $\sum_0^\infty f_k$ converges uniformly on D.

Proof. Let $S_n(x) = \sum_{k=0}^n f_k(x)$. We show that $\{S_n\}$ is uniformly Cauchy on D. Let $\varepsilon > 0$. For $m < n$ we have

$$S_n(x) - S_m(x) = f_{m+1}(x) + \cdots + f_n(x),$$

so

$$|S_n(x) - S_m(x)| \leq M_{m+1} + \cdots + M_n.$$

Since the series of constants $\sum_{k=0}^\infty M_k$ converges by hypothesis, there exists an integer N such that if $n > m \geq N$,

$$M_{m+1} + \cdots + M_n < \varepsilon.$$

This implies that for $n > m \geq N$,

$$|S_n(x) - S_m(x)| < \varepsilon$$

for all $x \in D$. Thus the sequence $\{S_n\}$ is uniformly convergent on D; that is, the series $\sum_0^\infty f_k$ is uniformly convergent on D. ■

Example 9.17: Consider again the geometric series $1 + x + x^2 + \ldots$ on the interval $[-a,a]$, for any $0 < a < 1$ (as we did in Example 9.15). Then $|x^k| \leq a^k$ for every $k = 0,1,2\ldots$ and $x \in [-a,a]$. Since $\sum_{k=0}^\infty a^k$ converges, by the M-test the series $\sum_{k=0}^\infty x^k$ converges uniformly on $[-a,a]$. ◀

Example 9.18: Let us investigate the uniform convergence of the series

$$\sum_{k=1}^{\infty} \frac{\sin k\theta}{k^p}$$

for values of $p > 0$. The crudest estimate on the size of the terms in this series is obtained just by using the fact that the sine function never exceeds 1 in absolute value. Thus

$$\left| \frac{\sin k\theta}{k^p} \right| \leq \frac{1}{k^p} \quad \text{for all } \theta \in \mathbb{R}.$$

Since the series $\sum_{k=1}^{\infty} 1/k^p$ converges for $p > 1$, we obtain immediately by the M-test that our series converges uniformly (and absolutely) for all real θ provided $p > 1$. In particular, as we shall see in subsequent sections, this series represents a continuous function, one that could be integrated term by term in any bounded interval.

We seem to have been particularly successful here, but a closer look also reveals a limitation in the method. The series is also pointwise convergent for $0 < p \leq 1$ (use the Dirichlet test) for all values of θ, but it converges nonabsolutely. The M-test cannot be of any help in this situation since it can address only absolutely convergent series. ◄

Because of the remark at the end of this example, it is perhaps best to conclude, when using the M-test, that the series tested "converges absolutely and uniformly" on the set given. This serves, too, to remind us to use a different method for checking uniform convergence of nonabsolutely convergent series (see the next section).

9.3.3 Abel's Test for Uniform Convergence

✂ *Advanced section. May be omitted.*

The M-test is a highly useful tool for checking the uniform convergence of a series. By its nature, though, it clearly applies only to absolutely convergent series. For a more delicate test that will apply to some nonabsolutely convergent series we should search through our methods in Chapter 3 for tests that handled nonabsolute convergence. Two of these, the Dirichlet test and Abel's test, can be modified so as to give uniform convergence.

A number of nineteenth century authors (including Abel, Dirichlet, Dedekind, and du Bois-Reymond) arrived at similar tests for uniform convergence. We recall that Abel's test for convergence of a series $\sum_{k=1}^{\infty} a_k b_k$ required the sequence $\{b_k\}$ to be convergent and monotone and for the series $\sum_{k=1}^{\infty} a_k$ to converge. Dirichlet's variant weakened the latter requirement so that $\sum_{k=1}^{\infty} a_k$ had bounded partial sums but required of the sequence $\{b_k\}$ that it converge monotonically to zero. Here we seek similar conditions on a series

$$\sum_{k=1}^{\infty} a_k(x) b_k(x)$$

of functions in order to obtain uniform convergence. The next theorem is one variant; others can be found in the Exercises.

Theorem 9.19 (Abel) *Let $\{a_k\}$ and $\{b_k\}$ be sequences of functions on a set $E \subset \mathbb{R}$.*

Suppose that there is a number M so that

$$-M \leq s_N(x) = \sum_{k=1}^{N} a_k(x) \leq M$$

for all $x \in E$ and every $N \in \mathbb{N}$. Suppose that the sequence of functions $\{b_k\} \to 0$ converges monotonically to zero at each point and that this convergence is uniform on E. Then the series

$$\sum_{k=1}^{\infty} a_k b_k$$

converges uniformly on E.

Proof. We will use the Cauchy criterion applied to the series to obtain uniform convergence. We may assume that the $b_k(x)$ are nonnegative and decrease to zero. Let $\varepsilon > 0$. We need to estimate the sum

$$\left| \sum_{k=m}^{n} a_k(x) b_k(x) \right| \tag{5}$$

for large n and m and all $x \in E$. Since the sequence of functions $\{b_k\}$ converges uniformly to zero on E, we can find an integer N so that for all $k \geq N$ and all $x \in E$

$$0 \leq b_k(x) \leq \frac{\varepsilon}{2M}.$$

The key to estimating the sum (5), now, is the summation by parts formula that we have used earlier (see Section 3.2). This is just the elementary identity

$$\sum_{k=m}^{n} a_k b_k = \sum_{k=m}^{n} (s_k - s_{k-1}) b_k$$

$$= s_m(b_m - b_{m+1}) + s_{m+1}(b_{m+1} - b_{m+2}) \cdots + s_{n-1}(b_{n-1} - b_n) + s_n b_n.$$

This provides us with

$$\left| \sum_{k=m}^{n} a_k(x) b_k(x) \right| \leq 2M \left(\sup_{x \in E} |b_m(x)| \right) < \varepsilon$$

for all $n \geq m \geq N$ and all $x \in E$ which is exactly the Cauchy criterion for the series and proves the theorem. ■

It is worth pointing out that in many applications of this theorem the sequence $\{b_k\}$ can be taken as a sequence of numbers, in which case the statement and the conditions that need to be checked are simpler.

Corollary 9.20: *Let $\{a_k\}$ be a sequence of functions on a set $E \subset \mathbb{R}$. Suppose that there is a number M so that*

$$\left| \sum_{k=1}^{N} a_k(x) \right| \leq M$$

for all $x \in E$ and every integer N. Suppose that the sequence of real numbers $\{b_k\}$ converges monotonically to zero. Then the series

$$\sum_{k=1}^{\infty} b_k a_k$$

converges uniformly on E.

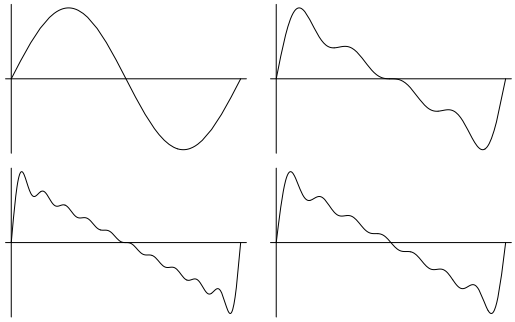


Figure 9.7. Graph of $\sum_{k=1}^{n} (\sin k\theta)/k$ on $[0, 2\pi]$ for, clockwise from upper left, $n = 1$, 4, 7, and 10.

Proof. Consider that $\{b_k\}$ is a sequence of constant functions on E and then apply the theorem. ■

In the exercises there are several other variants of Theorem 9.19, all with similar proofs and all of which have similar applications.

Example 9.21: As an interesting application of Theorem 9.19, consider a series that arises in Fourier analysis:

$$\sum_{k=1}^{\infty} \frac{\sin k\theta}{k}.$$

It is possible by using Dirichlet's test (see Section 3.6.13) to prove that this series converges for all θ.

Questions about the uniform convergence of this series are intriguing. In Figure 9.7 we have given a graph of some of the partial sums of the series.

The behavior near $\theta = 0$ is most curious. Apparently, if we can avoid that point (more precisely if we can stay a small distance away from that point) we should be able to obtain uniform convergence. Theorem 9.19 will provide a proof. We apply that theorem with $b_k(\theta) = 1/k$ and $a_k(\theta) = \sin k\theta$. All that is required is to obtain an estimate for the sums

$$\left| \sum_{k=1}^{n} \sin k\theta \right|$$

for all n and all θ in an appropriate set. Let $0 < \eta < \pi/2$ and consider making this estimate on the interval $[\eta, 2\pi - \eta]$. From Exercise 3.2.11 we obtain the formula

$$\sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta + \cdots + \sin n\theta = \frac{\cos\theta/2 - \cos(2n+1)\theta/2}{2\sin\theta/2}$$

and using this we can see that

$$\left| \sum_{k=1}^{n} \sin k\theta \right| \leq \frac{1}{\sin(\eta/2)}.$$

Now Theorem 9.19 immediately shows that

$$\sum_{k=1}^{\infty} \frac{\sin k\theta}{k}$$

converges uniformly on $[\eta, 2\pi - \eta]$.

Figure 9.7 illustrates graphically why the convergence cannot be expected to be uniform near to 0. A computation here is instructive. To check the Cauchy criterion on $[0, \pi]$ we need to show that the sums

$$\sup_{\theta \in [0,\pi]} \left| \sum_{k=m}^{n} \frac{\sin k\theta}{k} \right|$$

are small for large m, n. But in fact

$$\sup_{\theta \in [0,\pi]} \left| \sum_{k=m}^{2m} \frac{\sin k\theta}{k} \right| \geq \sum_{k=m}^{2m} \frac{\sin(k/2m)}{k} \geq \sum_{k=m}^{2m} \frac{\sin 1/2}{2m} > \frac{\sin 1/2}{2},$$

obtained by checking the value at points $\theta = 1/2m$. Since this is not arbitrarily small, the series cannot converge uniformly on $[0, \pi]$. ◀

Exercises

9.3.1 Examine the uniform limiting behavior of the sequence of functions

$$f_n(x) = \frac{x^n}{1 + x^n}.$$

On what sets can you determine uniform convergence?

9.3.2 Examine the uniform limiting behavior of the sequence of functions

$$f_n(x) = x^2 e^{-nx}.$$

On what sets can you determine uniform convergence? On what sets can you determine uniform convergence for the sequence of functions $n^2 f_n(x)$?

9.3.3 Prove that if $f_n \to f$ pointwise on a finite set D, then the convergence is uniform.

9.3.4 Prove that if $f_n \to f$ uniformly on a set E_1 and also on a set E_2, then $f_n \to f$ uniformly on $E_1 \cup E_2$.

9.3.5 Prove or disprove that if $f_n \to f$ uniformly on each set E_1, E_2, E_3, ..., then $f_n \to f$ uniformly on the union of all these sets $\bigcup_{k=1}^{\infty} E_k$.

9.3.6 Prove that if $f_n \to f$ uniformly on a set E, then $f_n \to f$ uniformly on every subset of E.

9.3.7 Prove or disprove that if $f_n \to f$ uniformly on each set $E \cap [a, b]$ for every interval $[a, b]$, then $f_n \to f$ uniformly on E.

9.3.8 Prove or disprove that if $f_n \to f$ uniformly on each closed interval $[a, b]$ contained in an open interval (c, d), then $f_n \to f$ uniformly on (c, d).

9.3.9 Prove that if $\{f_n\}$ and $\{g_n\}$ both converge uniformly on a set D, then so too does the sequence $\{f_n + g_n\}$.

9.3.10 Prove or disprove that if $\{f_n\}$ and $\{g_n\}$ both converge uniformly on a set D, then so too does the sequence $\{f_n g_n\}$.

9.3.11 Prove or disprove that if f is a continuous function on $(-\infty,\infty)$, then

$$f(x+1/n) \to f(x)$$

uniformly on $(-\infty,\infty)$. (What extra condition, stronger than continuity, would work if not?)

9.3.12 Prove that $f_n \to f$ converges uniformly on D if and only if

$$\lim_n \sup_{x\in D} |f_n(x) - f(x)| = 0.$$

9.3.13 Show that a sequence of functions $\{f_n\}$ fails to converge to a function f uniformly on a set E if and only if there is some positive ε_0 so that a sequence $\{x_k\}$ of points in E and a subsequence $\{f_{n_k}\}$ can be found such that

$$\left|f_{n_k}(x_k) - f(x_k)\right| \geq \varepsilon_0.$$

9.3.14 Apply the criterion in the preceding exercise to show that the sequence $f_n(x) = x^n$ does not converge uniformly to zero on $(0,1)$.

9.3.15 Prove Theorem 9.12.

SEE NOTE 244

9.3.16 Verify that the geometric series $\sum_{k=0}^{\infty} x^k$, which converges pointwise on $(-1,1)$, does not converge uniformly there.

SEE NOTE 245

9.3.17 Do the same for the series obtained by differentiating the series in Exercise 9.3.16; that is, show that $\sum_{k=1}^{\infty} kx^{k-1}$ converges pointwise but not uniformly on $(-1,1)$. Show that this series does converge uniformly on every closed interval $[a,b]$ contained in $(-1,1)$.

9.3.18 Verify that the series

$$\sum_{k=1}^{\infty} \frac{\cos kx}{k^2}$$

converges uniformly on all of $\mathbb{R}$.

9.3.19 If $\{f_n\}$ is a sequence of functions converging uniformly on a set E to a function f, what conditions on the function g would allow you to conclude that $g \circ f_n$ converges uniformly on E to $g \circ f$?

9.3.20 Prove that the series $\displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k}$ converges uniformly on $[0,b]$ for every $b \in [0,1)$ but does not converge uniformly on $[0,1)$.

9.3.21 Prove that if $\sum_0^{\infty} f_k$ converges uniformly on a set D, then the sequence of terms $\{f_k\}$ converges uniformly on D.

9.3.22 A sequence of functions $\{f_n\}$ is said to be *uniformly bounded* on an interval $[a,b]$ if there is a number M so that

$$|f_n(x)| \leq M$$

for every n and also for every $x \in [a,b]$. Show that a uniformly convergent sequence $\{f_n\}$ of continuous functions on $[a,b]$ must be uniformly bounded. Show that the same statement would not be true for pointwise convergence.

9.3.23 Suppose that $f_n \to f$ on $(-\infty,+\infty)$. What conditions would allow you to compute that

$$\lim_{n\to\infty} f_n(x+1/n) = f(x)?$$

9.3.24 Suppose that $\{f_n\}$ is a sequence of continuous functions on the interval $[0,1]$ and that you know that $\{f_n\}$ converges uniformly on the set of rational numbers inside $[0,1]$. Can you conclude that $\{f_n\}$ uniformly on $[0,1]$? (Would this be true without the continuity assertion?)

9.3.25 Prove the following variant of the Weierstrass M-test: Let $\{f_k\}$ and $\{g_k\}$ be sequences of functions on a set $E \subset \mathbb{R}$. Suppose that $|f_k(x)| \leq g_k(x)$ for all k and $x \in E$ and that $\sum_{k=1}^{\infty} g_k$ converges uniformly on E. Then the series $\sum_{k=1}^{\infty} f_k$ converges uniformly on E.

9.3.26 Prove the following variant on Theorem 9.19: Let $\{a_k\}$ and $\{b_k\}$ be sequences of functions on a set $E \subset \mathbb{R}$. Suppose that $\sum_{k=1}^{\infty} a_k(x)$ converges uniformly on E. Suppose that $\{b_k\}$ is monotone for each $x \in E$ and uniformly bounded on E. Then the series $\sum_{k=1}^{\infty} a_k b_k$ converges uniformly on E.

9.3.27 Prove the following variant on Theorem 9.19: Let $\{a_k\}$ and $\{b_k\}$ be sequences of functions on a set $E \subset \mathbb{R}$. Suppose that there is a number M so that

$$\left| \sum_{k=1}^{N} a_k(x) \right| \leq M$$

for all $x \in E$ and every integer N. Suppose that

$$\sum_{k=1}^{\infty} |b_k - b_{k+1}|$$

converges uniformly on E and that $b_k \to 0$ uniformly on E. Then the series $\sum_{k=1}^{\infty} a_k b_k$ converges uniformly on E.

9.3.28 Prove the following variant on Theorem 9.19: Let $\{a_k\}$ and $\{b_k\}$ be sequences of functions on a set $E \subset \mathbb{R}$. Suppose that $\sum_{k=1}^{\infty} a_k$ converges uniformly on E. Suppose that the series

$$\sum_{k=1}^{\infty} |b_k - b_{k+1}|$$

has uniformly bounded partial sums on E. Suppose that the sequence of functions $\{b_k\}$ is uniformly bounded on E. Then the series $\sum_{k=1}^{\infty} a_k b_k$ converges uniformly on E.

9.3.29 Suppose that $\{f_n\}$ is a sequence of continuous functions on an interval $[a,b]$ converging uniformly to a function f on the open interval (a,b). If f is also continuous on $[a,b]$, show that the convergence is uniform on $[a,b]$.

9.3.30 Suppose that $\{f_n\}$ is a sequence of functions converging uniformly to zero on an interval $[a,b]$. Show that $\lim_{n\to\infty} f_n(x_n) = 0$ for every convergent sequence $\{x_n\}$ of points in $[a,b]$. Give an example to show that this statement may be false if $f_n \to 0$ merely pointwise.

9.3.31 Suppose that $\{f_n\}$ is a sequence of functions on an interval $[a,b]$ with the property that $\lim_{n\to\infty} f_n(x_n) = 0$ for every convergent sequence $\{x_n\}$ of points in $[a,b]$. Show that $\{f_n\}$ converges uniformly to zero on $[a,b]$.

9.4 Uniform Convergence and Continuity

We can now address the questions we asked at the beginning of this chapter. We begin with continuity. We know that the pointwise limit of a sequence of continuous functions need not be continuous. We now show that the *uniform* limit of a sequence of continuous functions must be continuous.

Theorem 9.22: *Let $\{f_n\}$ be a sequence of functions defined on an interval I, and let $x_0 \in I$. If the sequence $\{f_n\}$ converges uniformly to some function f on I and if each of the functions f_n is continuous at x_0, then the function f is also continuous at x_0. In particular, if each of the functions f_n is continuous on I, then so too is f.*

Proof. Let $\varepsilon > 0$. We must show there exists $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon$$

if $|x - x_0| < \delta$, $x \in I$. For each $x \in I$ we have

$$f(x) - f(x_0) = (f(x) - f_n(x)) + (f_n(x) - f_n(x_0)) + (f_n(x_0) - f(x_0)),$$

so

$$|f(x) - f(x_0)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)|. \tag{6}$$

Since $f_n \to f$ *uniformly*, there exists $N \in \mathbb{N}$ such that

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3} \tag{7}$$

for all $x \in I$ and all $n \geq N$. We infer from inequalities (6) and (7) that

$$|f(x) - f(x_0)| < |f_N(x) - f_N(x_0)| + \frac{2}{3}\varepsilon. \tag{8}$$

We now use the continuity of the function f_N. We choose $\delta > 0$ such that if $x \in I$ and $|x - x_0| < \delta$, then

$$|f_N(x) - f_N(x_0)| < \frac{\varepsilon}{3}. \tag{9}$$

Combining (8) and (9), we have

$$|f(x) - f(x_0)| < \frac{\varepsilon}{3} + \frac{2}{3}\varepsilon = \varepsilon,$$

for each $x \in I$ for which $|x - x_0| < \delta$, as was to be shown. ■

Note. Let us look a bit more closely at the proof of Theorem 9.22. We first obtained $N \in \mathbb{N}$ such that the function f_N approximated f closely (within $\varepsilon/3$) on all of I. This function f_N served as a "stepping stone" toward verifying the continuity of f at x_0. There are three small "steps" involved:

1. $|f_N(x) - f(x)|$ is small (*for all* $x \in I$) because of uniform convergence.
2. $|f_N(x) - f_N(x_0)|$ is small (for all x near x_0) because of the continuity of f_N.
3. $|f_N(x_0) - f(x_0)|$ is small because $\{f_n(x_0)\} \to f(x_0)$.

If we tried to imitate the proof under the assumption of pointwise convergence, the first of these steps would fail. You may wish to observe the failure by working Exercise 9.4.2.

Theorem 9.22 can be stated in terms of series. Recall that a series $\sum_1^\infty f_k$ converges uniformly to the function f on D if the sequence

$$\{S_n\} = \{\sum_{k=1}^{n} f_k\}$$

of partial sums converges uniformly to f on D.